The Incompleteness of Observation

[Status note, 2026-07-26: this companion overview is superseded by the book, `book/`, which now carries its content in maintained form, and is frozen at this revision — it will not be updated in subsequent releases, and where it and the book disagree, the book is authoritative. Two elements are for now available only here: the D-gauge completeness walkthrough (§5.3 below) and the mathematical summary tables of the appendix, pending their incorporation into the book.]

Why Physics' Biggest Contradiction Might Not Be a Contradiction at All

With Complete Mathematical Detail

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The Worst Prediction in Physics

Physics has two spectacularly successful theories. Quantum mechanics describes the behavior of atoms, particles, and light. General relativity describes gravity, space, and time. Each has been confirmed to extraordinary precision. They have never disagreed with any experiment.

They disagree with each other.

Ask quantum mechanics how much energy empty space contains and it gives you a staggering number: roughly 10^{113} joules per cubic meter. Ask general relativity the same question — read the answer off the expansion rate of the universe — and you get about 6×10^{-10} joules per cubic meter. The ratio is 10^{122}. For scale, the number of atoms in the observable universe is about 10^{80}. This is not a close call.

For decades, the assumption has been that something is deeply broken — that one or both calculations contain an error, and that finding the mistake will point the way to a unified theory of everything.

This paper argues the opposite. Neither calculation is wrong. They disagree because they *must*. In fact, that massive 10^{122} discrepancy isn't a failure at all. It is the strongest piece of existing evidence we have that quantum mechanics is not the fundamental bedrock of reality, but an emergent description forced upon us by our limited vantage point.

The argument is built from a chain of mathematical proofs, each feeding into the next. This document explains what the paper claims, walks through the logic of every major proof, and shows how they connect.

The Starting Point: Observation Exists

Every mathematical proof starts from assumptions, and this framework has exactly one. It doesn't mention quantum mechanics. It doesn't mention general relativity. It is:

Observation occurs.

An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. This is the cogito of Descartes — "I think, therefore I am" — made mathematically precise. It is the one empirical fact that cannot be doubted: if you are reading this sentence, observation is occurring.

The paper formalizes this as a definition:

Definition. An *observation* is a triple (S, φ, V): a total system S, a deterministic dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$ — a proper subsystem with finitely many distinguishable internal states, coupled to the complement $H = S \setminus V$ through φ.

This single sentence contains no physics. It is weaker than classical mechanics (no continuity, no Hamiltonian, no Lagrangian). A shuffled deck of cards satisfies it. A finite cellular automaton satisfies it. Any finite computation satisfies it. The paper shows that three structural lemmas follow from this definition alone:

A note on "exactly one." The framework's foundational work has since looked harder at what "observation occurs" actually assumes, and found it is most honestly stated as **two** axioms rather than one. The first — that a distinction occurs, that there is some this-rather-than-that — genuinely cannot be doubted. The second — that the distinction *recurs*, that there is more than one moment — is a real further assumption: a single frozen instant, however full of difference, contains no time, and time cannot be squeezed out of difference alone. The framework assumes both, and a no-go result shows the second does not follow from the first. This is a strength rather than a hedge: the foundation can point to exactly which single ingredient is assumed rather than indubitable. A further question is genuinely open: whether the observer's status as a *proper part* of a larger whole — its embeddedness — is itself a third such assumption, or instead follows from the first axiom. This turns on a subtle point about what "a distinction occurs" means, it is under review, and — importantly — it changes no prediction the framework makes: it bears only on how the foundation's assumptions are counted and described. The companion paper *Physics Modulo Gauge* develops the foundational structure in full, as its Part I; for the purposes of this overview, "observation occurs" can be read as that two-part (possibly three-part) foundation, and nothing below depends on the refinement.

Lemma 1 (Finiteness). *The observer has finitely many distinguishable internal states, so the visible configuration space $\mathcal{C}_V$ is finite, with a discreteness scale ϵ providing a finite minimal cell volume.* There's a smallest meaningful size ε — you can't resolve anything smaller. This means the configuration space is finite, not continuous. This matters because finite systems have a property infinite systems don't — they must eventually return to their starting state (Poincaré recurrence). In Part I, ε is left unspecified. In Part II, self-consistency forces ε = 2 l_p (twice the Planck length).

Lemma 2 (Causal partition). *An observer is a proper subsystem $V \subsetneq S$. The complement $H = S \setminus V$ is the hidden sector.* The total phase space splits into two pieces:

$$\Gamma = \Gamma_V \times \Gamma_H$$

Γ_V is the visible sector (what the observer can access). Γ_H is the hidden sector (what they cannot). The total Hamiltonian splits correspondingly:

$$H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

H_V governs the visible sector alone. H_H governs the hidden sector alone. H_int couples them — it's how the two sectors talk to each other. Without H_int, the two sectors would evolve independently and the observer would never feel the hidden sector's influence.

Lemma 3 (Unique measure). *The counting measure on S — assigning equal weight to each state — is the unique measure invariant under φ.* The observer uses standard Kolmogorov probability theory. No exotic probability theories, no negative probabilities, no quantum probability — just ordinary probability. This is what makes the result surprising: we're putting in classical probability and getting out quantum mechanics.

That's it. The precise claim is layered. At the finite observable-law level, the resulting stochastic law has an exact fixed-basis Hilbert/unitary/Born-form representation ($S \iff D \iff Q_{\text{fb}}$); when C4 readback holds, the OI sector additionally carries genuine predictive memory and, for a fixed finite recurrent representative, global indivisibility. The Schrödinger/unitary interpolation and the $p = |U|^2$ dictionary belong to that exact representation layer. What does **not** follow from the definition alone is that every coherent preparation, non-commuting instrument, and spatial composite is already one standard local quantum operational theory. That common operational lift remains the narrow frontier; in Bell experiments the OI coupling graph supplies causal cones, while exact causal locality additionally requires operational setting screening. The hidden-sector conditions are:

C1: Non-zero coupling (H_int $\neq$ 0). The visible and hidden sectors interact. Information flows between them. Without this, the observer's room is perfectly isolated — nothing interesting happens.

C2: Memory persistence (τ_S $\ll$ τ_B). The hidden sector's *coarse-grained mixing time* — the rate at which a stored record of visible history degrades —

is long compared with the accessible window. This is persistence, not slowness: fast hidden dynamics that conserve the record serve equally well, and a shift-register bath rotating five cells per visible step supports a maximal readback gap ([Main §1.3]). τ_S is the timescale of visible-sector processes; τ_B is the timescale of hidden-sector processes. This is the *opposite* of the usual assumption in physics. Normally, people assume the environment is fast and chaotic (a "heat bath" that quickly forgets everything). Here, the environment is slow and has a long memory. This is what makes the dynamics non-Markovian.

C3: Sufficient capacity ($\log_2$|C_H| $\geq$ I*). The hidden sector has enough distinguishable states to encode the conditional past-future information the observer actually sees — the per-process bound, not a crude "many more degrees of freedom" comparison. It follows from readback by data processing in any faithful realization, and framework realizations saturate it ([Main §1.3]).

The definition sets the stage. The conditions determine what kind of show plays on it. The next section explains why the cosmological horizon satisfies all three.

The Observer's Blind Spot

Light travels at a finite speed, and the universe has a finite age. Put those together and every observer has a horizon — a boundary beyond which no signal has had time to arrive. Everything beyond that boundary is ordinary physics: fields, particles, radiation. But it is structurally inaccessible. Not because our telescopes aren't good enough, but because the geometry of spacetime forbids it. No technology that obeys the speed of light can reach past it.

This means every observer in the universe is in the same epistemic situation: there are degrees of freedom — a vast number of them — that influence what you measure but that you can never track. When you write down the laws of physics for the things you *can* see, you're forced to average over everything you can't. You have to "trace out" the hidden sector.

Here's what that looks like concretely. The total system — visible plus hidden — is deterministic. If you knew the complete state, you could predict the future exactly. But you don't know the hidden part. You know the visible state is x, but there are many possible hidden states compatible with x, and each one sends x to a different visible future. Hidden state h_1 might send the particle left; hidden state h_2 might send it right. Since you can't tell which hidden state you're in, the best you can do is assign probabilities: average over all the possible hidden states, weighted by how likely each one is. The result is a set of *transition probabilities* — the chance that visible state x at time t_1 becomes visible state y at time t_2. You've gone from a deterministic system you can't fully see to a probabilistic one you can. That's a stochastic process, and it's the only description available to any observer who can't access the hidden sector.

The standard expectation is that this should produce something boring — classical, memoryless noise. And it would, if the hidden sector were fast and forgettable, like air molecules bouncing off a grain of pollen. Each kick is independent of the last. Physicists call this *Markovian* behavior.

But the hidden sector beyond the cosmological horizon is not like that. It differs in three specific ways, and the paper proves that their conjunction changes everything.

It's coupled. The horizon is not a static wall. Stress-energy conservation enforces continuous dynamical correlations across it. Matter crosses the horizon, and the horizon area adjusts in response to interior energy density. Information flows in both directions. (Condition C1.)

It's slow. The hidden sector's correlation time is set by the Hubble timescale — roughly 10^{17} seconds, the age of the universe. Any laboratory experiment operates on timescales of 10^{-15} seconds or shorter. The ratio is 10^{-32}. The hidden sector cannot "reset" between your measurements. Every correlation it picks up from one experiment is still there when the next one begins. This is the *opposite* of the standard Markovian regime, where the environment decorrelates fast. Here, it never decorrelates at all. (Condition C2.)

It's vast. The hidden sector has roughly 10^{122} independent degrees of freedom — the Bekenstein-Hawking entropy of the cosmological horizon. No experiment you could ever perform would appreciably disturb its state. Its memory never saturates. (Condition C3.)

A fast environment with vast capacity would wash out correlations (Markovian noise). A slow environment with limited capacity would eventually fill up and stop recording. What the observer needs is an environment that couples to it, retains the record long enough to be read, has the capacity to hold it, and feeds it back — and it is the feedback (C4) that does the work, the other three following from it or supporting it. Such an environment sustains the persistent, non-decomposable correlations the paper calls *P-indivisibility* — a technical term meaning the system's transition probabilities at different times cannot be broken into independent steps.

Partition-Relativity (§1.4)

This is the first real proof in the paper, and it's beautifully simple.

What it proves: The emergent description (what the observer sees) depends *only* on the partition — on which degrees of freedom are visible and which are hidden. Nothing else.

The formula:

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h)$$

Unpacking each symbol:

- **T_ij**: The probability of transitioning from visible state x_i to visible state x_j in the time interval from t_1 to t_2. This is what the observer measures.
- **(x_i, h)**: The complete state — visible part x_i, hidden part h.
- φ**_{t2-t1}**: The deterministic evolution. Takes the complete state at time t_1 and returns the complete state at time t_2. Uniquely determined by the definition.
- π**_V**: Projection onto the visible sector. Takes a complete state (x, h) and returns just x.
- δ**_{xj}[...]**: The Kronecker delta. Equals 1 if the visible part ended up at x_j, equals 0 otherwise.
- **d**μ**(h)**: Integration over all possible hidden states, weighted by the Liouville measure.

In plain English: For each possible hidden state h, check whether starting at (x_i, h) and evolving forward lands the visible part on x_j. Count up all the hidden states where this happens, weighted by how likely each hidden state is. The result is the probability of the transition x_i → x_j.

The proof: The formula has exactly three inputs: (1) the dynamics φ_t — fixed by the definition, (2) the partition (Γ_V, Γ_H) and projection π_V — fixed by Lemma 2, and (3) the measure μ — fixed by Lemma 3 (Liouville measure is the unique choice). Since inputs 1 and 3 are determined by the definition, the only free input is the partition. Therefore: everything about the emergent description depends only on the partition. QED.

Why the Liouville measure is unique: The observer needs a "prior" — a way to weight the hidden states. Liouville measure is the unique measure on phase space that is absolutely continuous (no point masses) and invariant under Hamiltonian flow. Any smooth initial distribution evolves toward it. Singular measures are excluded by Lemma 3's requirement of standard probability theory. The observer has no choice.

Emergent Stochasticity and the Slow-Bath Regime (§§2.1–2.2)

The total system is deterministic. If you knew both x and h, you'd know the future with certainty. But the observer knows only x. Different hidden states h send the same visible state x to different futures.

Example: visible state is "Heads." Hidden state could be any die value 1–6. If the die is 1 or 2, the dynamics flip the coin to Tails. If the die is 3–6, the coin stays at Heads. The observer doesn't know the die, so they see: P(Heads → Tails) = 2/6 = 1/3. The randomness is epistemic (from ignorance) not ontological (from fundamental indeterminacy).

In a normal "heat bath" scenario, the environment is fast and chaotic. It scrambles any information you write into it before you can read it back. This produces Markovian (memoryless) dynamics — each step is independent of previous steps.

C2 inverts this — the condition itself is persistence, the record surviving to readback, with a slow hidden sector our universe's comfortable route (a fast bath that conserves and routes the record serves equally). The hidden sector here is slow. When the visible sector interacts with it (writing information through H_int), the information stays there. At the next interaction, the hidden sector reads back what was written before. The observer sees history-dependent transition probabilities — what happens next depends on what happened before.

This is non-Markovian dynamics. It's the key ingredient that separates quantum mechanics from classical stochastic processes.

The P-Indivisibility Theorem (§2.3)

What it claims. If a deterministic system is split into a visible and hidden sector, and these sectors are genuinely coupled, then the visible sector's behavior *cannot* be a simple memoryless random process. It must exhibit P-indivisibility — a specific kind of built-in memory.

What "P-indivisible" means. A stochastic process is "P-divisible" if you can always find a valid transition matrix connecting any two time points. Mathematically: for any times t_1 < t_2 < t_3, there exists a stochastic matrix Λ such that:

$$T(t_3, t_1) = \Lambda(t_3, t_2) \cdot T(t_2, t_1)$$

where Λ has non-negative entries and rows summing to 1. "P-indivisible" means this fails — the "intermediate propagator" would need negative entries, which means it's not a valid probability matrix.

Breuer, Laine, and Piilo proved that P-indivisibility is equivalent to "information backflow" — the system's distinguishability can *increase* over time. In a classical Markov process, you can only lose information (mixing). In a P-indivisible process, information comes back. This is exactly what quantum systems do — interference, revivals, and non-classical correlations all involve information returning from where it was stored.

The setup. We work on finite sets (Lemma 1). The visible sector has states C_V = {x_1, x_2, …} with |C_V| $\geq$ 2. The hidden sector has states C_H = {h_1, h_2, …}. The total dynamics is a bijection φ on C_V $\times$ C_H. The transition matrix is:

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

The key tool — total variation distance:

$$d(p, q) = \frac{1}{2} \sum_k |p_k - q_k|$$

This measures how distinguishable two probability distributions are. If d = 1, they're perfectly distinguishable. If d = 0, they're identical. For P-divisible processes, d can only decrease or stay constant.

Step 1 — Recurrence. φ is a bijection on a finite set. Keep applying φ and you must eventually return to where you started — there are only finitely many states to visit. Formally: there exists N such that φ^N = id. So T^(N) = I, and:

$$d(\delta_i T^{(N)}, \delta_j T^{(N)}) = d(\delta_i, \delta_j) = 1$$

After N steps, states that started distinguishable are still perfectly distinguishable.

Step 2 — Strict contraction. Suppose T is not a permutation matrix. (This does not follow from C1, which asserts only non-zero coupling; it is the separate non-permutation witness the theorem assumes at this step.) So there exist states i, j, l where both T_il > 0 and T_jl > 0. The total variation distance after one step:

$$d(\delta_i T, \delta_j T) = \frac{1}{2} \sum_k |T_{ik} - T_{jk}| < 1$$

The inequality is strict because the distributions overlap. Distinguishability has decreased.

Step 3 — The punchline. At t = 1: d < 1 (distinguishability decreased). At t = N: d = 1 (distinguishability restored). The distinguishability went down then came back up — non-monotonic behavior. A P-divisible process can only have non-increasing distinguishability. Therefore the process is P-indivisible. QED.

The proof uses almost nothing — just that the dynamics is a bijection on a finite set (the definition and Lemma 1) and that the coupling is non-trivial (C1). It is purely combinatorial.

The Accessible-Timescale Lemma (§2.3 continued)

The recurrence proof shows P-indivisibility exists as a mathematical property. But the recurrence time is absurdly long — for the cosmological case, it's e$^{(10}$122) years. Nobody will ever observe it.

The accessible-timescale lemma shows that information backflow happens on *laboratory timescales*, independently of recurrence.

The mechanism: At each interaction (timescale τ_S), the coupling H_int transfers some information from the visible sector to the hidden sector. Call the amount I_0. Between interactions, the hidden sector's correlations decay with a rate set by its spectral gap Δ ~ 1/τ_B.

The decay per visible-sector step is:

$$e^{-\Delta\tau_S} \approx 1 - \frac{\tau_S}{\tau_B}$$

When τ_S ≪ τ_B (C2), this is very close to 1 — almost no decay. The hidden sector remembers almost perfectly between steps. After k steps, the cumulative decay is:

$$e^{-k\Delta\tau_S} \approx 1 - \frac{k\tau_S}{\tau_B}$$

As long as k · τ_S ≪ τ_B, the hidden sector retains ~k bits of visible-sector history. The mutual information satisfies:

$$I(X_{<t}; X_{>t} \mid X_t) \geq I_0 \left(1 - \frac{k\tau_S}{\tau_B}\right)$$

For the cosmological case: τ_S ~ 10^{-15} s, τ_B ~ 10^{17} s. Even after k = 10^{20} steps, k · τ_S/τ_B ~ 10^{-12} — negligible. The hidden sector remembers everything.

The role of C3: The hidden sector's memory capacity is log_2(|C_H|) bits. If k bits of history are written but the capacity is only m < k bits, old data gets overwritten. C3 ensures m is large enough that the memory never saturates on observable timescales.

The Coin-and-Die Model (§2.4)

The paper builds a concrete toy model to make the mechanism tangible.

Setup: - Visible: x ∈ {0, 1} (a coin: 0 = Heads, 1 = Tails) - Hidden: h ∈ {1, 2, 3, 4, 5, 6} (a die) - Total: 12 states

The permutation σ:

Input state	Output state	What happens
(0, 1)	(1, 1)	Coin flips, die stays
(1, 1)	(0, 1)	Coin flips, die stays
(0, 2)	(1, 2)	Coin flips, die stays
(1, 2)	(0, 2)	Coin flips, die stays
(0, 3)	(0, 4)	Coin stays, die changes
(0, 4)	(0, 3)	Coin stays, die changes
(0, 5)	(0, 6)	Coin stays, die changes
(0, 6)	(0, 5)	Coin stays, die changes
(1, 3)	(1, 4)	Coin stays, die changes
(1, 4)	(1, 3)	Coin stays, die changes
(1, 5)	(1, 6)	Coin stays, die changes
(1, 6)	(1, 5)	Coin stays, die changes

Every swap is a transposition (a ↔ b), so σ^2 = id (apply twice, everything returns).

Checking the conditions: C1 (coupling): die values 1 and 2 flip the coin ✓. C2 (memory persistence): σ^2 = id means recurrence time is 2 steps, giving τ_S/τ_B = 1/2 ✓. C3 (sufficient capacity): 6 hidden states vs 2 visible states ✓.

Computing T(1,0). Start at x = 0 (Heads). All 6 die values are equally likely.

- h = 1: σ(0,1) = (1,1) → Tails
- h = 2: σ(0,2) = (1,2) → Tails
- h = 3: σ(0,3) = (0,4) → Heads
- h = 4: σ(0,4) = (0,3) → Heads
- h = 5: σ(0,5) = (0,6) → Heads
- h = 6: σ(0,6) = (0,5) → Heads

P(0 → 0) = 4/6 = 2/3, P(0 → 1) = 2/6 = 1/3. By the same logic for x = 1:

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

Distinguishability at t = 1:

$$d(\delta_0 T, \delta_1 T) = \frac{1}{2}(|2/3 - 1/3| + |1/3 - 2/3|) = 1/3$$

Started at d = 1. Now d = 1/3. Distinguishability decreased.

What Markov would predict at t = 2: Apply the same transition matrix again:

$$T(1,0)^2 = \begin{pmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{pmatrix}$$

Distinguishability would drop to d = 1/9. More mixing.

What actually happens at t = 2: $\sigma^2 = \mathrm{id}$. Every state returns to its starting point.

$$T(2,0) = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Distinguishability is back to d = 1. Complete un-mixing. Impossible for a Markov process.

The smoking gun — negative entries. If there were a valid stochastic matrix Λ(2,1) connecting steps 1 and 2:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = I \cdot \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The entries -1 are negative. No valid stochastic matrix exists. **This is P-indivisibility.**

The mechanism in detail. The die works as a memory register. At step 1, if the coin was at 0 and the die was at 1, the coin flips to 1 but the die stays at 1. The die value 1 now encodes the information "the coin was at 0 and I flipped it." At step 2, σ sees (1, 1) and flips it back to (0, 1). The die read its own memory and reversed the flip. C1 (coupling) allows writing to the memory. C2 (memory persistence) ensures it isn't erased between reads. C3 (sufficient capacity) ensures there's enough room. C4 (history readback): at step 2 the unchanged die register controls the reverse flip — the dynamics reads back the record written at step 1. In this toy, C1 through C4 explain the memory mechanism — the information backflow; the explicit negative-propagator calculation above independently establishes the stronger P-indivisibility witness, and the process is thereby, via the stochastic-quantum correspondence, the carrier of a nontrivial quantum representation in compressed form under (T).

Why conditions C2 and C3 matter physically

The P-indivisibility theorem needs only coupling (C1) and finiteness. So why does the paper insist on slow memory (C2) and vast capacity (C3)?

Because P-indivisibility without C2 and C3 might only show up at absurd timescales or might self-destruct. C2 ensures the memory persists on timescales accessible to actual experiments, not just at cosmic recurrence times. C3 ensures the hidden sector never runs out of room to store information — if it saturates, later imprints overwrite earlier ones, and the process becomes effectively memoryless. Together with C4's readback — the load-bearing condition (`review4_probes.py`: storage and capacity without readback do not suffice) — C2 and C3 make the accessible memory strong, persistent, and observationally relevant; the P-indivisibility witness is the stronger diagnostic that readback makes measurable.

The Stochastic-Quantum Correspondence (§3.1 and Appendix A)

This is the key link. Section 2 proved that the embedded observer's dynamics are P-indivisible. Section 3 shows how the correspondence supplies the quantum representation for exactly this sector — admitted by every finite law, nontrivial here.

The core statement. Any P-indivisible stochastic process on a finite configuration space of size n can be embedded into a unitarily evolving quantum system. Specifically, there exists a Hilbert space H (dimension $\leq$ n^3) and a unitary operator U(t) such that:

$$T_{ij}(t) = \sum_{\alpha} |\tilde{U}_{(i,0)\,(j,\alpha)}(t)|^2 \qquad \text{(ancilla-marginal; the bare visible form is the } m_a = 1 \text{ special case)}$$

This is the Born rule. The left side is the transition probability computed by averaging over hidden states (the classical formula from partition-relativity). The right side is the quantum mechanical probability — the squared modulus of a matrix element of the unitary evolution operator. The equivalence is not approximate. It is not an analogy. It is a mathematical identity.

Two routes to the same layer — and the in-house one is now primary. The imported route uses Barandes' stochastic-quantum correspondence (2023–2025), which supplies the compressed form: P-indivisibility means transition probabilities can't be factored through intermediate times — try it and you get "negative probabilities." In quantum mechanics, this is *exactly what happens*: probability amplitudes combine to produce interference patterns that don't factorize classically. What Barandes proved is that every finite-configuration process in his class — a broad one, including Markov chains — can be regarded as a subsystem of a unistochastic process, so the two descriptions are notational variants of one object. The breadth matters: the representation's existence is not the framework's distinctive result, and what P-indivisibility adds is that the dynamics so represented is genuinely indivisible.

The in-house route, given in Appendix A, uses Stinespring's dilation theorem (1955), and it is now primary — its permutation unitary is exactly the object behind the equivalence theorem's internal $D \Rightarrow Q_{\mathrm{fb}}$ leg ([Main §3.4]): a deterministic bijection on a finite product space defines a permutation unitary; tracing out the hidden sector with the Liouville measure produces a completely positive quantum channel whose diagonal elements recover the classical transition probabilities exactly. This second route requires only textbook results. Both reach the transition-statistics layer and neither reaches the operational instrument algebra, which stays open; together they ensure that *layer* rests on no single recent result.

Where the quantum features come from:

- **The Schrödinger equation** arises because U(t) is differentiable. Any continuous one-parameter *group* of unitaries — U(s+t) = U(s)U(t) — can be written as U(t) = exp(-iHt/$\hbar$) for one fixed Hermitian H (Stone's theorem); the framework's realized processes supply exactly such a group at horizon times (the fixed-Ĥ lemma), while a general smooth family without the group property would need a time-dependent generator. Feynman famously remarked of this equation in Volume III, Lecture 16: "Where did we get that equation from? Nowhere. It is not possible to derive it from anything you know. It came out of the mind of Schrödinger." The derivation above is the answer that wasn't available in 1963. The equation is no longer "from nowhere" — it is the differential form of unitary evolution, and unitary evolution is available, under (T), for embedded observers' processes — nontrivially so where the dynamics is P-indivisible by C1–C4.
- **The Born-form dictionary** is the ancilla-marginal rule $T_{ij} = \sum_\alpha |\widetilde{U}_{(i,0)(j,\alpha)}|^2$ in general, reducing to $T_{ij} = |U_{ij}|^2$ only in the trivial-ancilla case. Inside Q_{fb} that quadratic readout is part of the definition of the equivalent representation. The permutation-unitary construction does not, however, select the exponent: its dilated matrix entries are 0 or 1, so replacing the square by $|\widetilde{U}|^p$ leaves the same marginal table for every $p > 0$. Why the quadratic functional is the physical dictionary is therefore a question about the dictionary's origin, not a gap in the fixed-basis equivalence.
- **The action scale** $\hbar$ enters when converting from the dimensionless unitary to a dimensionful Hamiltonian: Ĥ = i$\hbar$ (∂U/∂t) U†. The value of $\hbar$ cannot be determined from the dimensionless transition data alone — it requires additional physical input from the partition geometry (§5).

- **Bell correlations.** Bare non-factorization is not enough to select the quantum correlation set. For spatial OI the dependency graph supplies exact causal cones; with operational setting screening and an indivisible joint law, the process lies in the causal-local class for which the CHSH/Tsirelson result applies. Without the local-setting premise, the same broad reversible machinery can realize post-quantum finite behaviors, so the Bell statement is explicitly conditional.

The Phase-Locking Lemma (§3.1 continued)

A potential objection: the relation T_ij = |U_ij|^2 throws away phase information. Different unitaries could give the same transition probabilities. Does this make the quantum description ambiguous?

The phase-locking lemma shows: no.

Setup: The transition probability at time t is:

$$T_{ij}(t) = \left| \sum_k V_{ik}\, e^{-iE_k t}\, V_{jk}^* \right|^2$$

where V_ik = ⟨i|k⟩ are the overlaps between the configuration basis and the energy eigenbasis, and E_k are the energy eigenvalues. Expanding the square:

$$T_{ij}(t) = \sum_{k,l} V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*\; e^{-i(E_k - E_l)t}$$

The Fourier trick: This is a sum of oscillating terms at frequencies ω_kl = E_k - E_l. If all these frequencies are distinct (condition G2: non-degenerate energy gaps), you can extract each coefficient by Fourier transform:

$$a_{ij}^{kl} = V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*$$

Extracting the moduli: Setting i = j: $a_{ii}^{kl} = |V_{ik}|^2 |V_{il}|^2$. If none of the overlaps are zero (condition G3), all moduli |V_ik| are determined.

Extracting the phases: Write V_ik = |V_ik| e^{iφ_ik}. The argument of the Fourier coefficient gives:

$$\arg(a_{ij}^{kl}) = (\varphi_{ik} - \varphi_{il}) - (\varphi_{jk} - \varphi_{jl})$$

The only transformation preserving all double differences is φ_ik → φ_ik + α_i + β_k — just relabeling (choosing a different phase convention for the basis states). Once you fix these conventions, all remaining phases are determined up to one further discrete ambiguity: the complex-conjugate Hamiltonian H $\to$ –H*, an antiunitary twin producing identical transition probabilities at every time.

Bottom line: Continuous-time transition probability data determines the Hamiltonian up to relabeling conventions and the antiunitary conjugate H $\to$ –H* (ancilla-case completeness open beyond tested generic cases); i.e. up to physically irrelevant relabeling.

Bell Inequality Violations (§3.2)

This is the question everyone asks: isn't this ruled out by Bell's theorem?

What Bell's theorem actually requires. For this framework the crucial distinction is between *dynamical causal propagation* and *statistical setting independence*. Full spatial OI supplies the first: because the coupling graph is defined by one-step dependence, a local state after k updates can depend only on the radius-k graph ball. A remote setting therefore cannot physically propagate into the local state before the causal cones meet.

That fact alone does not imply Bell's statistical locality conditions. If the setting variables are correlated with preparation-relevant hidden data, conditioning on a setting can reweight the local hidden ensemble without any superluminal signal. Exact Bell causal locality therefore needs an additional **operational setting-screening** condition. This is a genuine open quantitative obligation of the cosmological/mixing realization, not something supplied by C1–C4 alone.

Why the unrestricted OI class is broader. A globally controlled reversible realization can couple both wings through the joint history and even let a wing update depend on the remote control. Such models are outside the local-setting premise and can realize PR-box behavior. Their existence is the negative control showing that no-signaling, generic non-factorization, or P-indivisibility by themselves select the quantum correlation set.

Jarrett/Fine scope. Parameter independence, outcome independence, and Fine's theorem are useful diagnostics, but they are not a derivation of the quantum set here. In particular, graph causality should not be silently promoted to parameter independence when setting-conditioned hidden ensembles differ.

The Tsirelson result. The applicable result is the stronger one: for a **causally local indivisible** stochastic process, the CHSH value obeys the Tsirelson bound $2\sqrt{2}$. Thus the OI route is: graph causal cones + operational setting screening + indivisibility $\Rightarrow$ causal-local class $\Rightarrow$ Tsirelson. With imperfect screening, Main gives the corresponding quantitative robustness bound in terms of $I_{\mathrm{MI}} = I(\Lambda; XY)$.

The Characterization Theorem (§3.3)

It's not enough to show that embedded observation *produces* QM (sufficiency). The paper shows the observed non-Markovian dynamics *require* embedded observation under C1–C4 (necessity, per condition; the C2 leg conditional on the mixing hypothesis). The full logical chain:

- Barandes proved: an exact correspondence between quantum systems and the source's indivisible class, defined at the level of first-order conditionals (the framework's link to it is the translation hypothesis (T), [Main §3.1])
- Section 2.3 proved: a non-permutation coupling $\implies$ P-indivisibility at the recurrence scale, and C4's readback $\implies$ accessible non-Markovianity, quantitatively (sufficiency)
- Section 3.3 proves: P-indivisibility $\implies$ C1, C3, C4 (necessity, unconditional); C2's slow form within the mixing class (the conditional physical-memory theorem)
- Combined: **accessible non-Markovianity $\iff$ embedded observation under C1, C3, C4 (per finite horizon) — C2 is not needed for the equivalence and carries its physical content separately, through the conditional physical-memory theorem — with a fixed-basis unitary quantum representation internal and universal ([Main §3.4]), the compressed form supplied across the correspondence's class under (T), nontrivially where the dynamics is P-indivisible**

Necessity of C1 (coupling). With no hidden influence at any accessible step, every visible transition is a fixed function of the visible present — the process is Markov, hence not the observed one. A non-permutation one-step matrix is the cheapest *witness* of C1, sufficient but not necessary: influence can hide at step one and surface later (the delayed-revival family).

Necessity of C2 (memory persistence), the slow form conditional on mixing. Between coupling events (separated by τ_S), the hidden sector evolves under its own Hamiltonian. The convergence to equilibrium is:

$$\|e^{\mathcal{L}_H \tau_S} \mu_H(\cdot|x_i) - \mu_{\mathrm{eq}}\|_{\mathrm{TV}} \leq C\, e^{-\Delta \tau_S}$$

In the fast-bath regime ($\Delta \cdot \tau$_S $\gg$ 1), this is exponentially small. The hidden sector forgets everything between interactions. Each transition is computed against the same equilibrium distribution, so T^(k) = T^k — a Markov chain, hence P-divisible. Contrapositive, within the mixing class: P-indivisibility requires τ_S $\ll$ τ_B — conditional because the convergence bound presupposes a relaxation gap Δ. Fast *non-mixing* hidden dynamics — conserved circulation — retain records with no timescale separation at all (`fastbath_probes.py`, [Main §1.3]), which is why (C2) is stated in two forms and this necessity belongs to the slow form.

Necessity of C3 (sufficient capacity). The non-Markovian mutual information is bounded by the hidden sector's size:

$$I(X_{<t}; X_{>t} \mid X_t) \le \log_2 m$$

where m = |C_H|. Proof: the total system is deterministic, so X_{>t} is a function of (X_t, H_t). Given X_t, the chain X_{<t} → H_t → X_{>t} is Markov. By the data processing inequality:

$$I(X_{<t}; X_{>t} \mid X_t) \le I(X_{<t}; H_t \mid X_t) \le H(H_t \mid X_t) \le H(H_t) \le \log_2 m$$

Each step uses a standard information-theoretic inequality. The result: if you want K bits of history dependence, you need m ≥ 2^K hidden states.

The complete characterization. For |C_V| ≥ 2: the process is non-Markovian on accessible timescales ⟺ it arises, on each finite horizon, from marginalizing a deterministic bijection under C1, C3, C4 (C2 sits outside the equivalence: the existential realization satisfies it trivially with a static bath). Both directions are the foundational paper's own theorems; every finite-horizon law admits the fixed-basis unitary quantum representation ($S \iff D \iff Q_{\text{fb}}$, internal — the compressed form for the correspondence's class under (T)), and the fixed-Ĥ form is constructive for every realized process. The old three-way biconditional is retired: representability alone does not force memory (diagonal Hamiltonians), and conditional memory alone does not force P-indivisibility (the XOR family).

What "unitarily evolving QM" means precisely. At the finite observable-law layer the framework gives an exact fixed-basis Hilbert/unitary/Born representation: $S \iff D \iff Q_{\text{fb}}$. This is not metaphorical, and it includes exact multi-time fixed-basis records through the permutation-unitary realization. It does **not** by itself derive the whole operational category of quantum mechanics. The visible–hidden tensor factor used by the dilation is part of the representation construction; for two spatial laboratories, the OI dependency graph supplies exact causal cones, but a common local tensor/composite instrument structure is part of the coherent operational lift still stated separately in Main §3.4. Conditioning on a registered classical outcome is represented by ordinary conditioning on the substratum ensemble; identifying every coherent quantum update with that construction is likewise part of the operational lift. The memory-bearing sector ($M_t > 0$), and on a fixed finite representative the associated global indivisibility, carries the specifically OI content.

A note on what is proved versus imported, to keep the claim honest. The exact finite-law representation is internal: every finite stochastic law has a reversible deterministic realization and an exact fixed-basis unitary/Born representation ($S \iff D \iff Q_{\text{fb}}$, [Main §3.4]). That statement is universal — ordinary Markov chains are included — so representability itself is not the discriminator.

C4 supplies the nontrivial content: accessible history readback, unavoidable hidden predictive memory, and, on one fixed finite recurrent representative, global indivisibility. The Barandes stochastic–quantum correspondence remains useful for a compressed representation and its interpretive apparatus; it is not needed to prove that some Hilbert/Born representation exists. The remaining foundational qualification is narrower than the old "selection" wording: the framework has not yet proved that all coherent preparations, noncommuting instruments, and composites generated by one OI system form one standard local quantum operational theory. Spatial OI supplies graph causal cones, while Bell causal locality additionally requires operational setting screening. Thus the accurate headline is strong but layered: the finite observable stochastic/reversible/quantum descriptions are exactly equivalent; readback identifies the memory-bearing indivisible OI sector; the common coherent local operational lift is stated separately rather than silently folded into the equivalence theorem.

The Cosmological Application (§4)

4.1 The Partition

The cosmological horizon is the boundary beyond which no signal traveling at or below c can ever reach the observer. In a universe with a positive cosmological constant (de Sitter space), this horizon exists for every observer and has a definite, finite area:

$$A = \frac{4\pi c^2}{H^2}$$

where H is the Hubble parameter. This implements Lemma 2 naturally: Γ_V = everything inside the horizon, Γ_H = everything beyond.

Why $\hbar$ is the same for all observers: Different observers have slightly different horizons. But the gap equation $\hbar$ = c$^3\varepsilon^2$/(4G) depends only on local geometric quantities (c, G, ε) — not on the horizon area or the observer's worldline. So all observers derive the same $\hbar$.

4.2 Verification of the Conditions

C1 (coupling). In general relativity's ADM formulation, the bulk Hamiltonian is a sum of constraints that vanish on-shell — meaning the "real" dynamics happens at the boundary. The Hamiltonian and momentum constraints correlate interior and exterior data. This is stronger than just H_int ≠ 0: the constraints enforce correlations that persist on all timescales.

C2 (memory persistence). τ_B ~ 1/H ~ 10^{17} seconds (the Hubble time). τ_S ~ 10^{-15} seconds (a typical atomic process). The ratio: τ_S/τ_B ~ 10^{-32}.

C3 (sufficient capacity). The hidden sector has $A/\varepsilon^2 \sim 10^{122}$ modes (the de Sitter entropy). The visible sector has $\sim 10^{80}$ baryons. No experiment comes close to saturating the hidden sector.

4.3 Application

With the cosmological horizon satisfying the definition and all conditions, the characterization theorem applies. The observer's reduced description is non-Markovian — P-indivisible in the recurrence regimes — and admits a unitarily evolving quantum representation ([Main §3.4]). The value of $\hbar$ is determined by the partition geometry — which is what Section 5 derives.

Where Planck's Constant Comes From (§5)

Partition-relativity proved that the emergent quantum description is completely and uniquely determined by the partition. This means $\hbar$ cannot be a free parameter — it must be fixed by the geometry of the boundary.

The Classical Horizon Temperature (§5.1)

Starting point: Jacobson's identity. Jacobson (1995) showed that Einstein's field equations follow from applying the first law of thermodynamics $\delta Q = TdS$ to local causal horizons:

$$dE = \frac{c^2\kappa}{8\pi G}\,dA$$

where κ is the surface gravity and dA is the change in horizon area. This is a classical gravitational identity — no quantum mechanics involved.

The entropy density is $\eta = 1/\varepsilon^2$ — one coupled mode per minimal cell of area ε^2. This is not an assumption about the number of states per cell: ε is defined as the minimal distinguishable scale (Lemma 1), so each cell of area ε^2 contributes exactly one boundary mode that couples across the partition. The number of internal states per mode (the alphabet size q) is a gauge freedom with no observable consequences [SM §2.7]. So $dS = dA/\varepsilon^2$. From $dE = TdS$:

$$k_B T_{\mathrm{cl}} = \frac{c^2\epsilon^2\kappa}{8\pi G}$$

For the de Sitter horizon, $\kappa = cH$:

$$k_B T_{\mathrm{cl}} = \frac{c^3\epsilon^2 H}{8\pi G}$$

Critical point: no $\hbar$ appears anywhere. This temperature is computed from purely classical quantities.

The Four-Step Derivation (§5.2)

Step 1 (Uniqueness). Partition-relativity guarantees $\hbar$ is determined by the partition geometry. It's not a free parameter.

Step 2 (Boundary-only dependence). A substantial lemma showing that $\hbar$ depends only on the boundary modes, not on the deep interior of the hidden sector. Decompose the hidden sector into boundary modes C_B (near the horizon) and deep modes C_D (far from the horizon):

$$\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$$

The proof has three parts: (i) spatial locality means V talks to B, and B talks to D, but V doesn't talk directly to D; (ii) on timescales t $\ll$ τ_B, the deep sector barely moves (displacement ~ 10^{-32}); (iii) because the deep modes are frozen, the sum over them produces a trivial factor:

$$T_{ij}(t) = \underbrace{\frac{1}{|\mathcal{C}_B|} \sum_b \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i, b))]}_{T_{ij}^{(B)}(t)} + \mathcal{O}(t/\tau_B)$$

The transition probabilities depend only on boundary dynamics. Since $\hbar$ is determined by transition probabilities, $\hbar$ depends only on boundary quantities: c, G, and ε. This excludes dependence on H — if $\hbar$ depended on H, observers at different cosmic epochs would have different quantum mechanics.

Step 3 (Dimensional analysis). Step 2 excludes volumetric (deep-sector) quantities, leaving boundary quantities. The boundary carries both *local* geometric data (ε, κ, and the constants c, G) and a *global* quantity: the total area A, which forms the dimensionless ratio A/ε^2 = S_dS. If $\hbar$ depended on S_dS, it would be observer-dependent — different observers have different horizon areas — contradicting the universality of the emergent action scale. This excludes A. The surface gravity κ is excluded by two independent arguments: (i) κ differs between horizon types (cosmological vs. black hole), but $\hbar$ is universal — a laboratory experiment measures the same $\hbar$ regardless of which horizon defines the partition; (ii) for the cosmological horizon, κ = cH is time-dependent ($\dot{H}$ $\neq 0$), but $\hbar$ is observed to be constant on laboratory timescales. The unique combination of c, G, and ε with dimensions of action:

$$[\hbar] = \frac{[c]^3\,[\epsilon]^2}{[G]} = \text{kg} \cdot \text{m}^2/\text{s} \quad \checkmark$$

So $\hbar = \beta\, c^3\varepsilon^2/G$, where β is a dimensionless constant that dimensional analysis alone can't fix.

Step 4 (Thermal self-consistency). We have two independent descriptions of the horizon temperature:

Classical: T_cl = $c^2\varepsilon^2\kappa/(8\pi G$k_B) — computed from the geometric substratum, no $\hbar$.

Quantum: The emergent QFT lives on a spacetime with a bifurcate Killing horizon. Standard QFT on curved spacetime gives a KMS thermal state at temperature:

$$T_Q = \frac{\hbar\kappa}{2\pi c k_B}$$

The two temperatures are computed independently — $T_{\rm cl}$ from the classical substratum alone (no QM), T_Q from the emergent QFT alone (no classical substratum details) — but they describe the same physical degrees of freedom: the boundary modes across which the partition is defined. Since the quantum description is derived from the classical one (Part I), and the derivation is exact at the boundary, the two descriptions cannot assign contradictory temperatures. Consistency requires $T_{\rm cl} = T_Q$:

$$\frac{c^2\epsilon^2\kappa}{8\pi G} = \frac{\hbar\kappa}{2\pi c}$$

The surface gravity κ cancels from both sides. Solving:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}}$$

This fixes $\beta = 1/4$.

Why this is not circular. The KMS temperature T_Q contains $\hbar$ as an *unknown.* The classical temperature T_cl contains no $\hbar$ at all. The non-circularity is structural: Part I establishes that a QFT emerges with *some* action scale $\hbar$; §5 determines *which* $\hbar$, using the independent classical temperature that Part I neither requires nor produces. If T_cl had depended on the deep hidden-sector volume (it doesn't — the boundary-only lemma excludes it), or if T_Q had been state-dependent (it isn't — the KMS temperature is purely kinematic), the matching wouldn't work. That neither pathology obtains makes the gap equation a genuine determination.

Predictive content. The gap equation relates one free parameter (ε) to one output ($\hbar$), but the derivation has genuine content because it picks out the specific relationship $\hbar = c^3\varepsilon^2/(4G)$ — rather than any other function of c, G,

ε — as the one that produces the Bekenstein-Hawking factor 1/4 (§6), the CC dissolution (§7.3), and the Type II RVM dark-energy form (§8.1) simultaneously. Any alternative $\hbar(\varepsilon)$ would fail at least one check. See Appendix item 1 for the detailed discussion.

The D-Gauge Completeness Theorem (§5.3)

The problem. The relation T_ij = |U_ij|^2 discards phase information. Could different Hamiltonians give the same transition probabilities?

The theorem. If |U'_ij(t)|^2 = |U_ij(t)|^2 for all i, j, t, then H' = DHD† where D is a diagonal unitary (a physically meaningless relabeling of basis phases).

The proof in three steps:

Step 1 (eigenvalue recovery): Fourier analysis of T_ij(t) extracts the energy differences E_k - E_l. Non-degeneracy of energy gaps means E'_k = E_k + E_0 (same eigenvalues up to a global shift).

Step 2 (modulus recovery): The diagonal Fourier coefficients give |V_ik|^2 directly. So |V'_ik| = |V_ik|.

Step 3 (phase structure): Writing V'_ik = V_ik e^{iδ_ik} and requiring all Fourier coefficients to match gives the double-difference condition:

$$\delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl} = 0 \pmod{2\pi}$$

The general solution: δ_ik = α_i + β_k — a sum of a row phase and a column phase. This is just basis rephasing.

The dimensional obstruction. The unitary U(t) is dimensionless. The Hamiltonian Ĥ = i$\hbar$ ∂_tU · U† contains $\hbar$, which is dimensionful. No amount of dimensionless data can fix a dimensionful constant. This is why Step 4 (thermal matching) is not just a convenient check but the *mathematically obligatory* step: it's the only place where dimensionful physical input enters the framework.

The Discreteness Scale (§6)

What ε = 2 l_p means

Rearranging the gap equation $\hbar$ = c$^3\varepsilon^2$/(4G):

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\, l_p^2$$

where l_p = √($\hbar$G/c^3) is the Planck length. Therefore ε = 2 l_p.

What this is: The framework has one free geometric parameter — the discreteness scale ε. The gap equation fixes its relationship to $\hbar$. Given the measured value of $\hbar$, the discreteness scale is determined: $\varepsilon = 2$ l_p $\approx 3.2 \times 10^{-35}$ meters.

What this is NOT: This is not an independent prediction of ε. The framework contains one free parameter and one equation relating it to known constants. The self-consistency condition pins ε to the Planck regime but doesn't predict a number that wasn't already implicitly known.

The Bekenstein-Hawking Entropy — Why the 1/4 factor is derived

The number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\,l_p^2}$$

This is the Bekenstein-Hawking entropy:

$$S_{\text{BH}} = \frac{A}{4\,l_p^2}$$

The factor of 1/4 — which Bekenstein and Hawking introduced as a proportionality constant in 1973 — is here derived: each minimal cell of area $\varepsilon^2 = 4$ l_p^2 contributes one unit of entropy. The 4 in the denominator comes from $\varepsilon = 2$ l_p.

Why this is significant: The 1/4 factor has been a mystery for 50 years. Most frameworks either assume it or derive it within constructions specifically designed to produce it. Here it follows from the gap equation with no additional input.

Self-consistency bounds on ε

If $\varepsilon^2 \ll$ l_p^2: Sub-Planckian cells would need further subdivision, creating a second trace-out within the first. This would make $\hbar$ multi-valued — contradicting the observed universality of $\hbar$.

If $\varepsilon^2 \gg$ l_p^2: Super-Planckian cells would be too coarse. The emergent quantum description would assign distinct quantum states to configurations that are physically identical.

The self-consistency condition $\varepsilon = 2$ l_p is the unique value where neither pathology occurs.

The Two Levels

Now the cosmological constant problem dissolves. But first, a crucial link: how does finite-dimensional quantum mechanics become quantum *field* theory — with independent modes, each carrying zero-point energy?

The answer is spatial locality. The classical substratum has it — neighboring cells interact, distant cells don't. The paper proves that the emergent quantum description inherits it. The argument is direct: the Barandes correspondence maps each classical configuration $(x_1, x_2, \ldots, x_N)$ to a quantum basis state $|x_1, x_2, \ldots, x_N\rangle$. That's already a tensor product. The only question is whether the emergent Hamiltonian respects the structure. And it must, because if two configurations differ only at sites that aren't neighbors, the classical dynamics can't connect them in an infinitesimal time step — so the transition probability between them is zero — so the corresponding quantum Hamiltonian matrix element is zero. The emergent Hamiltonian couples only neighboring cells, exactly like the classical one. You get a lattice quantum field theory with a built-in ultraviolet cutoff at $\epsilon = 2\, l_p$.

This is what makes the CC dissolution work. The QFT has independent modes. Each mode gets a zero-point energy. The sum diverges. And the "worst prediction in physics" follows — but only within the emergent description.

The critical insight is that general relativity and quantum field theory are not competing answers to the same question. They are answers to *different questions*, asked at different levels of the same reality.

Level 1: The classical substratum. Spacetime geometry is part of the fundamental layer. The metric tensor evolves via Einstein's field equations. The stress-energy tensor that sources gravity is the classical stress-energy of the total microstate. At this level, the vacuum energy density sits at the critical scale: $\rho \sim H^2/G \sim 10^{-9}$ J/m^3. No zero-point energy. No discrepancy.

Level 2: The emergent quantum description. For an embedded observer tracing out the hidden sector, the mandatory quantum description assigns a zero-point energy of $\frac{1}{2}\hbar\omega$ per mode. Sum to the Planck cutoff and you get $\rho_{\text{QFT}} \sim 10^{113}$ J/m^3. This number is real *within the quantum description* — it reflects the magnitude of the trace-out noise — but it is not a source term in Einstein's equations, because those equations operate at the classical level, which is logically prior to the quantum description.

This ordering is not a choice but is forced by three independent requirements. First, the partition must be definite — not in superposition — for the trace-out to be well-defined; a partition in superposition would yield an incoherent mixture of inequivalent quantum theories. Second, the partition is defined by the causal structure, which is determined by null geodesics of the metric; if the metric were derived from QM, the derivation would be circular (QM $\to$ metric $\to$ causal structure $\to$ partition $\to$ QM). Third, $\hbar$ is determined by the boundary geometry; if the geometry were itself quantum-mechanical, $\hbar$ would depend on a quantum state, contradicting its observed universality.

The 10^{122} ratio between the two answers is not a discrepancy. It equals S_{dS} — the Bekenstein-Hawking entropy of the cosmological horizon — which is the number of hidden-sector degrees of freedom the trace-out compresses into the emergent quantum state. The "worst prediction in physics" is the information compression ratio of the observer's blind spot. A category error, not a fine-tuning failure.

This is not a prediction awaiting future data. The observed vacuum energy has been measured since 1998, and it sits exactly at the classical geometric scale — the value the framework expects. Meanwhile, every attempt to search for a cancellation mechanism within quantum-first frameworks have found nothing. The framework explains why: there is nothing to cancel.

This track record is itself evidence for the ordering. The question of whether geometry is prior to quantum mechanics or quantum mechanics is prior to geometry is usually treated as a philosophical preference. But it has an empirical signature sitting in plain sight: one ordering produces the worst prediction in physics and has no solution; the other predicts the observed value and explains the discrepancy as a derived quantity. That doesn't prove the ordering is correct, but it's existing evidence, not a future hope. (Note the division of labor with the non-adjudicability result in the gravity paper: two descriptions that fit the same observations equally well can never be ranked from inside — that is the factoring argument — but the two *programs* here are not in that situation. One predicts the observed vacuum energy and one misses it by 10^{122}; predictive asymmetry on accessible data is exactly the kind of criterion the factoring argument leaves open.)

What Quantum Weirdness Looks Like From Here

Feynman opens his quantum mechanics volume with a famous warning. The double-slit experiment, he writes in Volume III, Lecture 1 of the *Lectures on Physics*, is "a phenomenon which is impossible, absolutely impossible, to explain in any classical way, and which has in it the heart of quantum mechanics. In reality, it contains the only mystery." He goes on: "We cannot make the mystery go away by 'explaining' how it works. We will just tell you how it works."

If quantum mechanics is an emergent description forced on embedded observers, the standard quantum puzzles acquire straightforward readings — and the "only mystery" stops being mysterious. It becomes a structural consequence.

The double-slit experiment. The particle goes through one slit. The interference pattern arises because the transition probabilities are computed by marginalizing over the hidden sector, and the hidden sector includes the field configuration near both slits. Opening or closing the second slit changes the boundary conditions, which changes the marginalization, which changes the pattern. The "wave-like" behavior is the hidden sector's influence shifting when the geometry changes. The mystery dissolves not into another postulate but into a derivation: this is what marginal statistics look like when paths through the partition are indistinguishable.

Entanglement. Two particles prepared together share a joint transition matrix inherited from the trace-out. The correlations are encoded in the structure of the dynamics itself, not in a hidden variable you could integrate over. This is why Bell inequality violations occur: the standard factorization assumption fails for indivisible processes. The framework reproduces quantum correlations exactly up to Tsirelson's bound, without faster-than-light signaling and without superdeterminism.

The measurement problem. Measurement produces definite outcomes through the indivisible dynamics. When Wigner can't access his Friend's lab, he traces out its internal degrees of freedom and assigns a superposition. The superposition reflects Wigner's epistemic situation — what *he* can infer — not the Friend's physical state. Branching in the Many-Worlds sense is a feature of the compressed description, not of the underlying reality.

Schrödinger's cat. The cat is always alive or always dead — the substratum holds exactly one configuration at every moment, and nothing in the box is smeared. The superposition is the *outside observer's* ledger: the record of which outcome occurred has been written into degrees of freedom the observer cannot read, so the honest description from outside prices both branches, the way the coin-and-die model (§2.4) prices its unobserved intermediate step. And the famous absurdity dissolves on a second look: a real cat is never an unobserved step. Air molecules, thermal photons, the Geiger counter itself — the environment reads the record continuously, and each interaction forces the statistics back to ordinary either/or probabilities. That is why no consciousness was ever needed to open the box; one stray photon is observer enough. Schrödinger built the scenario as a reductio — he thought a theory implying alive-and-dead cats had gone wrong somewhere — and the framework agrees, while relocating the error: not in the dynamics, which needed no collapse law added, but in reading the observer's compression algorithm as the state of the cat.

The common thread across the three readings is that what looked like irreducible quantum mystery is the partition's reflection of substratum dynamics. You're embedded in the universe; you don't get the view from nowhere; the description you arrive at is constrained by what marginal access permits. Feynman taught his students that the rules of QM "had to just be accepted." The framework's claim is that the rules don't have to be accepted — they have to be derived, and the derivation makes contact with the fact of embedding. Whether the derivation is correct is a separate question; the empirical predictions that distinguish it from competing accounts are listed below.

Predictions

The framework is not just a reinterpretation. It makes specific, falsifiable predictions — and, crucially, it makes predictions in domains it was not designed to address.

Dark energy evolution. Because the hidden sector's dimensionality changes as the Hubble parameter H evolves (the horizon area is $A = 4\pi c^2/H^2$), the emergent vacuum energy inherits a dependence on H. The predicted form matches the Running Vacuum Model: $\Lambda_{\text{eff}} = \Lambda_0 + \nu H^2$. The framework's *structural* commitments — boundary entropy depends on H, matter is conserved in the trace-out, and Bianchi consistency then forces $G(H)$ to run in tandem with $\rho_\Lambda(H)$ — fix the Type II RVM functional form. The magnitude of ν is fixed at theorem level by the emergent-QFT calculation: applying standard curved-spacetime running to the Standard Model fields at their physical masses on the derived classical background yields $|\nu_{\text{QFT}}| \sim (M_{\text{SM}}/M_{\text{Pl}})^2 \sim 10^{-32}$ — observationally indistinguishable from zero. A potential additional contribution from the substratum mode count is constrained by the boundary architecture: modes are catalogued by spatial location alone (one mode per cell of area ϵ^2, no inherent frequency assignment), so a natural 2D distribution concentrates them near the UV cutoff and contributes well below $|\nu_{\text{QFT}}|$. Absent additional substratum structure giving a log-uniform boundary-mode distribution at the cosmological horizon, the framework's prediction reduces to $|\nu| \approx |\nu_{\text{QFT}}|$. A direct test is provided by Bertini, Novaes, von Marttens, and Shapiro (arXiv:2509.24026), who fit a renormalization-group-corrected ΛCDM model — structurally equivalent to Type II RVM at leading order, with both $G(\mu)$ and $\Lambda(\mu)$ running and matter conserved — against Planck 2018 + DESI DR2 + DES-Y5, reporting $\nu = -(2.5 \pm 1.3) \times 10^{-4}$, consistent with zero at 2σ and consistent with the framework's prediction. The *structural* prediction — Type II dynamical DE in running-vacuum form — stands as a parameter-free framework commitment. DESI Year 5 ($\sim 1\%$ precision on ν) is the decisive test: a null at that precision confirms the prediction; a high-significance detection of $|\nu|$ at the 10^{-4} level would require additional substratum structure beyond what the present boundary-entropy construction supplies — structure a direct computation of the boundary spectral measure from the substratum dynamics now shows the linear map does not generate (the measure is power-law local, never log-uniform), so such a detection would also demand its correlated Lorentz-violation co-signature.

The dark sector as corroboration. The trace-out that produces the memory-bearing dynamics quantum mechanics represents has a gravitational consequence that the paper did not set out to find. The boundary entropy — the S_{dS} modes traced out to produce the quantum description — has thermal energy that, distributed over the Hubble volume, equals the critical density exactly. A crucial subtlety: this thermal energy is computed entirely from pre-trace-out (classical) quantities — the number of boundary modes, the classical horizon temperature, and the Hubble volume — with no reference to $\hbar$ or the emergent quantum description. This is what distinguishes the boundary entropy's gravitational contribution from the QFT zero-point energy ($\rho_{\mathrm{QFT}} \sim 10^{113}$ J/m^3), which exists only *after* the trace-out and is an artifact of the emergent description. The framework denies that the zero-point energy gravitates (this is the CC dissolution of Part II); the boundary entropy's classical thermal energy *does* gravitate, at the scale $\rho_{\mathrm{crit}} \sim 10^{-9}$ J/m^3.

The paper proves that this entropy has no operator in the emergent QFT. The baryonic sector — what QFT can account for — is ~5% of ρ_{crit}. The remaining ~95% is the boundary entropy: gravitationally active, invisible to the emergent description, and persistent through P-indivisibility (condition C2). This matches the observed composition of the universe, in which ~95% of the gravitational content has no source in particle physics. The uniform component corresponds to dark energy (handled by Part II's CC dissolution). The structured component — dark matter — arises from matter-induced entropy displacement: baryonic matter displaces boundary entropy via the Clausius relation, the Jacobson mechanism converts the entropy gradient into curvature, yielding the MOND acceleration scale $a_0 = cH/6 \approx 1.2 \times 10^{-10}$ m/s^2 and the baryonic Tully-Fisher relation $v^4 = GM_B \cdot cH/6$ — both parameter-free (GR, §7.3). That a definition designed to formalize observation also account for the dark sector's total budget and internal structure is independent corroboration that observational incompleteness is capturing real structure.

High-redshift dark matter. Because $a_0(z) = cH(z)/6$ and $H(z)$ increases with redshift, the dark matter phenomenology evolves: a_0 is 1.8× larger at $z = 1$, 3.0× at $z = 2$, and 4.6× at $z = 3$. This shrinks the MOND crossover radius, making galaxies more baryon-dominated at high redshift — their rotation curves should decline beyond a smaller radius. Genzel et al. (Nature, 2017) report exactly this: stacked rotation curves at $z = 0.9$–2.4 show declining outer velocities at $> 3\sigma$ significance relative to local spirals. The cleanest local test is even more striking: Jiao et al. (*A&A* 678, A208, 2023) detect for the first time a Keplerian decline in the Milky Way's own rotation curve from ~ 19 to ~ 26.5 kpc using Gaia DR3 kinematics, with the flat rotation curve hypothesis rejected at 3σ. Our own galaxy is now the strongest single piece of evidence for the $H(z)$-dependent crossover prediction at $z = 0$ — the framework predicts the crossover at $r_M \sim 17$ kpc for the Milky Way's baryonic mass, essentially where Jiao et al. observe the transition. The baryonic Tully-Fisher relation also evolves: $v_{\mathrm{flat}} \propto H(z)^{1/4}$, predicting 32% higher velocities at $z = 2$ at fixed baryonic mass. McGaugh et al. (2024) report no evolution in the *stellar* mass TF to $z \sim 2.5$ — but this is actually *predicted* by the framework, because gas fractions at high z are large ($f_{\mathrm{gas}} \sim 50$–70%) and the gas mass omitted from M_* almost exactly compensates the dynamical shift (the cancellation gas fractions — 44% at $z = 1$, 67% at $z = 2$ — match observations). The definitive test is the *baryonic* TF at $z > 1$ with reliable ALMA gas masses. Particle dark matter (NFW halos) predicts flat rotation curves at all redshifts — the observed decline is unexpected in ΛCDM but natural in the OI framework. Direct dark matter searches continue to return null results (LZ Collaboration's 417-day analysis presented December 2025 is the most sensitive WIMP search ever conducted, finding nothing), consistent with the framework's prediction that no particle dark matter exists.

Cluster scales and the Bullet Cluster. Galaxy clusters — the hardest test for any MOND-like theory — are addressed by the interpolation between the Newtonian and deep-MOND regimes. The simple interpolation function $g_{\text{total}} = g_B \cdot \nu(g_B/a_0)$ with $\nu(y) = (1 + \sqrt{1 + 4/y})/2$ matches the Coma cluster to $< 1\%$ in velocity (1260 vs 1270 km/s) and reduces the standard MOND mass shortfall from a factor ~ 2 to ~ 1.0–1.5 for other rich clusters — with the residual attributable to undetected warm-hot intergalactic medium (WHIM). This interpolation is indistinguishable from the deep-MOND limit at galaxy scales (differences < 0.07 dex, well within the observed RAR scatter). The Bullet Cluster — where gravitational lensing peaks at the galaxy positions rather than the dominant X-ray gas — is explained by the non-local character of entropy displacement: the boundary entropy relaxation time is $\sim H^{-1} \approx 14$ Gyr, while the collision crossing time is ~ 0.15 Gyr. The dark gravity is frozen at the pre-collision configuration (centered on the galaxies, which defined the potential wells for gigayears), not tracking the recently displaced gas. This reproduces the observed lensing morphology and makes a testable prediction: very old post-collision systems should show gradual relaxation of the dark gravity toward the gas distribution (GR, §7.3). The same thermodynamic averaging explains why the entropy displacement reproduces the CMB acoustic peak pattern: oscillating perturbations have zero net entropy displacement per cycle (the Clausius relation involves *net* heat transfer), so only the growing mode is tracked — providing non-oscillating potential wells identical to CDM in the linear regime.

Neutrino predictions and the JUNO + DESI record. The framework predicts Majorana neutrinos with normal mass ordering and a hierarchical spectrum, with Σm_ν near the oscillation minimum of 0.059 eV. The DESI DR2 + CMB analysis (Elbers et al., March 2025) reports $\Sigma m_\nu < 0.0642$ eV (95% CL), prefers normal ordering, and bounds the lightest neutrino mass at $m_l <$ 0.023 eV — every directly comparable measurement matches the OI prediction. The same analysis finds a 3σ tension with the lower oscillation limit assuming ΛCDM, interpreted as "a hint of new physics not necessarily related to neutrinos" — exactly the structural mismatch that the OI dark energy resolves. JUNO first results (November 2025) deliver world-leading precision on $\sin^2\theta_{12} = 0.3092 \pm 0.0087$, matching the OI prediction $1/3 - 1/(4\pi^2) = 0.3080$ at 0.14σ against the JUNO direct measurement. The post-JUNO global fit of Capozzi et al. (2025) tightens the world average to 0.3085 ± 0.0073, against which the OI prediction matches at 0.07σ. JUNO's measurement also confirms the persistent solar/reactor 1.5σ tension. The neutrino sector and the dark energy sector are coupled in the framework — both follow from the same lattice structure — so the joint observational support is not independent confirmation of separate effects but a single empirical pattern consistent with the framework's central mechanism.

Gauge coupling prediction. The companion paper [SM §6] extends the derivation chain to the gauge coupling strengths. The fermion-induced coupling gives $1/\alpha_0 = 23.25$ at the Planck scale — a universal value determined by the lattice structure ($N_f = 6$ flavors, $T(R) = 1/2$), not by the specific bijection φ. Combined with non-perturbative gauge self-energy corrections (from pure-gauge Monte Carlo at the induced coupling) and Standard Model renormalization group running, this reproduces all three SM gauge couplings at M_Z: $1/\alpha_1 = 59.00$, $1/\alpha_2 = 29.57$, $1/\alpha_3 = 8.47$ — matching the observed values to $< 0.1\%$.

Independent corroboration of the trace-out mechanism. The framework's central technical move — that integrating out hidden degrees of freedom from a deterministic substrate produces well-defined non-Markovian visible dynamics, which under the Barandes correspondence becomes quantum mechanics — is now established at theorem level in the open systems literature. Brandner (*Phys. Rev. Lett.* **134**, 037101 and companion *Phys. Rev. E* **111**, 014137, January 2025) proves that for autonomous linear evolution equations, integrating out inaccessible degrees of freedom yields well-defined non-Markovian visible dynamics in a controlled weak-memory regime, with explicit error bounds and a convergent perturbation scheme — derived independently for general open systems with no stake in OI being right. Direct experimental demonstrations exist in controlled physical systems: Mehl et al. (PRL 108, 220601, 2012) showed that hidden slow degrees of freedom in a colloidal system produce non-Markovian visible dynamics violating naive Markovian fluctuation theorems; Gröblacher et al. (Nature Communications 6, 7606, 2015) observed non-Markovian Brownian motion in a macroscopic micromechanical oscillator. On the quantum side, Kim (April 2025) shows that monitored quantum systems are formally quantum hidden Markov models with a rigorous correspondence to classical hidden Markov models — the same formal structure as the stochastic-quantum bridge, developed independently in the monitored quantum systems literature. The framework's central technical move is therefore not speculative; it is the standard way physicists now think about open systems with hidden structure, and the mathematical apparatus has been published in the standard literature within the past 12 months.

No competing framework produces all of these from a single definition. The parallel with the cosmological constant dissolution is exact: the 10^{122} discrepancy is the *information compression ratio* of the trace-out, and the ~95% dark sector is the *gravitational occlusion fraction*. Together, they account for the two largest anomalies in modern cosmology as two aspects of a single phenomenon: the cost of observing the universe from within.

Philosophical Lineage

The paper is a physics paper, but its core claims — that observers face irreducible limits, that two irreconcilable descriptions can both be correct, that incompleteness is a structural feature rather than a deficiency — sit at the intersection of some of the oldest debates in philosophy. A systematic mapping against the major traditions reveals a striking pattern: broad support for most of the framework, and near-universal resistance to one specific thesis.

The seven claims

The framework rests on seven implicit philosophical commitments:

1. **Embedded observers face irreducible limits.** No observer inside a system can access the complete state.
2. **QM and GR are both correct** within their domains.
3. **The hidden sector is permanently inaccessible** — not due to technological limitations, but structural ones.
4. **The underlying reality is local and definite.** Indeterminacy belongs to the observer's description, not to the world.
5. **Incompleteness is structural, not deficient.** The limitation arises because the observer is made of the same elements as the universe it is trying to describe — a physical form of self-reference. This is analogous to Gödel's incompleteness theorem, not to ignorance that better instruments could cure.
6. **The description is observer-relative.** Different partitions yield different emergent physics.
7. **The two descriptions are irreconcilable** — not because one is wrong, but because they are complementary projections of a single reality that no embedded observer can access directly.

Claims 1, 5, 6, and 7 enjoy broad philosophical support across nearly every tradition examined. Claim 4 — that the underlying reality is definite — is the paper's most philosophically isolated thesis.

The Gödel connection

The analogy between this framework and Gödel's incompleteness theorem is not merely metaphorical. Gödel proved that a formal system rich enough to encode arithmetic cannot prove all true statements about itself from within — the limitation arises because the system is self-referential, capable of constructing sentences that refer to its own provability. The observer in this framework faces a structurally parallel situation: the reason a cosmological horizon exists at all is that the observer is a physical subsystem made of the same fields, obeying the same speed-of-light constraint, as the universe it is trying to describe. An observer not made of the universe's own elements would not face a horizon and would not be forced into a quantum description. The incompleteness is a consequence of self-inclusion.

The connection can be tightened through Wolpert's limits of inference [19], which the paper cites — with one distinction kept explicit, because collapsing it is precisely the misuse this section exists to avoid. Wolpert proved, using diagonal self-referential arguments directly descended from Gödel's, that any inference device embedded in the universe it is trying to predict faces fundamental limits — not because of noise or finite resources, but because complete self-prediction is logically impossible. That is a *logical* incompleteness: worst-case, unconditional, and unquantified — the diagonal exhibits one question the device must fail, and by itself yields no probability calculus, no Hilbert space, and no entropy budget. The framework's incompleteness is a *physical* one that runs parallel to it rather than instantiating it: the causal partition enforces the trace-out, the trace-out produces the reduced law — P-indivisible in the memory-bearing sector the conditions characterize — and the representation is universal, the imported correspondence supplying the compressed form across its class, P-indivisibility fixing whether it is nontrivial under the translation hypothesis (T). The two come apart cleanly — an observer in a universe without dark energy would be fully subject to Wolpert's theorem yet would face no cosmological horizon — and the 10^{122} compression ratio (the Bekenstein-Hawking entropy of the horizon) measures the *physical* incompleteness, which is contingent on the observed causal structure and has no counterpart in the purely logical results. A closer rigorous relative of the physical statement is T. Breuer's theorem (1995, *Philosophy of Science* 62: 197–214; not to be confused with H.-P. Breuer of the Breuer–Laine–Piilo indivisibility criterion cited elsewhere in this framework) that no measurement performed from inside a system can perfectly resolve the state of a whole that contains the measuring apparatus; Wolpert and Breuer secure, each in its own domain, the rigor of the *pattern* that the framework's construction then realizes physically.

The key difference from Gödel is in the *form* of self-reference. Gödel's is syntactic: the system encodes a sentence that says "I am not provable." The framework's is physical: the observer is made of the described, so complete description would require the observer to fully encode its own state plus everything causally connected to it, which the causal structure forbids. Wolpert sits between the two, using Gödelian logic applied to physical systems. Together they form a family of results sharing one architecture — not a derivation in which each implies the next: Gödel (formal systems cannot completely describe themselves), Wolpert (physical inference devices cannot completely predict the systems they inhabit), this framework (an embedded observer's description is forced through a trace-out whose cost is the quantum form of the reduced law). Each stands on its own proof in its own domain; none is a corollary of the others, and the framework's physics would survive unchanged had the logical results never been proven — they corroborate the pattern, they do not supply the mechanism. One assumption in Wolpert's theorems also deserves naming rather than burying: they quantify over all possible setups of the inference device, a counterfactual-availability assumption whose status in a deterministic substratum is exactly that of the free-choice assumption in Bell's theorem. The framework's treatment is the same one it applies to Bell-type assumptions generally — identify the descriptive level at which the assumption holds (the observer's effective description, where the framework reproduces the represented quantum statistics exactly — the fixed-basis transition layer) as distinct from the substratum level, where it need not.

There is also a difference worth stating plainly, because it sets the standard the framework should be held to. Gödel's and Turing's results are complete proofs from minimal definitions — they assume only the formal system itself and smuggle in nothing about the world. The incompleteness here is of the same *architecture* (a limitation that follows from observing a system you are part of) but not yet of the same *self-containment*: the framework's quantitative results lean on commitments that are read from the observed world (that space is three-dimensional, that the lattice is cubic) rather than forced by the logic of self-inclusion alone. The honest reading is that the framework meets the Gödel–Turing standard for the part that is genuinely forced — the limitation theorem and its representation-theoretic consequences — while the empirically-anchored inputs have no counterpart in those purely formal results. Naming that boundary is what keeps the analogy a structural argument rather than borrowed grandeur.

Hofstadter's *Gödel, Escher, Bach* argues that self-referential systems produce genuinely emergent higher levels — "strange loops" where a system's hierarchical levels fold back on themselves, generating properties invisible at the lower level. This framework agrees: quantum mechanics is the real, irreducible description available to any embedded observer, just as Hofstadter's "I" is real even though it emerges from neurons. The strange loop is intact — the observer, made of the universe's own elements, generates through self-inclusion an emergent description that governs everything the observer can access. The disagreement with Hofstadter is narrow but consequential: he argues that once the emergent level is established, the substrate beneath it is explanatorily inert. The cosmological constant problem suggests otherwise. The 10^{122} discrepancy exists precisely because the emergent quantum description and the classical substrate assign different vacuum energies, and only the substrate's value matches observation. The loop has a floor, and the floor matters.

Closest ancestor: Nicholas of Cusa

Of all thinkers surveyed, the fifteenth-century cardinal Nicholas of Cusa provides the most precise structural alignment. His *docta ignorantia* (learned ignorance) is essentially Claim 5: the highest knowledge is knowing what we cannot know, and this is an intellectual achievement, not a failure. His *coincidentia oppositorum* (coincidence of opposites) maps onto Claim 7 with remarkable precision: contradictions irreconcilable in the finite realm dissolve in infinity — the infinitely large circle's circumference becomes a straight line, the infinite polygon becomes a circle. QM and GR, irreconcilable within any finite observational framework, would be coincident in the infinite ground that generates both.

Most strikingly, Cusa's "wall of Paradise" from *De Visione Dei* maps onto Claim 3: an insurmountable boundary beyond which finite intellect cannot pass. Scholars like Emmanuel Falque interpret reaching this wall not as escaping through it but as *inhabiting the boundary* — compatible with a framework where understanding the limit is itself the deepest available insight.

Deepest ontological parallel: Spinoza

Spinoza's attribute theory is arguably the single closest ontological parallel. One substance (God/Nature) expresses itself through infinite attributes, of which humans know only two: Thought and Extension. Jonathan Bennett's "barrier doctrine" captures the key feature: each attribute must be conceived through itself — no explanatory flow crosses between them. This is precisely Claim 7: two complete, correct descriptions that are structurally irreconcilable, yet both describe the same underlying reality. Spinoza's substance is fully determinate, aligning with Claim 4. His parallelism doctrine — the order of ideas is the same as the order of things — means the two descriptions track the same structure through incommensurable vocabularies.

The divergence is epistemological: Spinoza believes reason achieves adequate knowledge of reality through *scientia intuitiva* (intuitive knowledge). The paper's permanent limits directly contradict this ambition.

The recurring fault line: Claim 4

Across every tradition examined, a striking pattern emerges. The claim that underlying reality is "local and definite" faces resistance from virtually every direction:

Kant prohibits positive characterization of the noumenal realm as dogmatic metaphysics — you can know *that* things-in-themselves exist but never *what* they are. **Hegel** diagnoses a performative contradiction: to posit a hidden sector and characterize it as containing standard physics is already to have crossed the boundary you claim is uncrossable. **Nietzsche** attacks it as residual Platonism — having correctly shown that all observation is perspectival, the paper reinstates the very "true world" his career was dedicated to destroying. **Wittgenstein** rejects it as nonsensical: asserting what lies beyond the limits of the sayable transgresses exactly the limits the paper identifies.

Nagarjuna identifies it as *svabhāva*-reification — attributing inherent existence to what Buddhist emptiness (*śūnyatā*) says lacks it. **Daoism** warns against naming the unnameable: the Dao that can be spoken is not the eternal Dao, and calling reality "definite" is itself an act of conceptual carving. **Whitehead** insists reality is fundamentally processual and creative, involving genuine indeterminacy — removing that indeterminacy robs the cosmos of its creative character.

Advaita Vedanta comes closest to full support — Brahman is indeed a definite underlying reality — but insists it is *accessible*: the observer IS the underlying reality, and liberation (*mokṣa*) consists in recognizing this identity. Every Hindu and Buddhist soteriology rejects permanent inaccessibility.

Only Spinoza (whose fully determinate substance is naturally definite) and Worrall's epistemic structural realism (which posits real but unknowable natures behind structures) provide genuine philosophical support for Claim 4.

What's genuinely new

The paper's philosophical lineage is not a single line of descent but a mosaic. Its structural epistemology draws from Kant through Wittgenstein to Wolpert. Its complementarity thesis combines Bohr's physics with Cusa's coincidence of opposites and Daoist yin-yang. Its projection metaphysics reaches through Plato's cave and Plotinus's emanation to Advaita Vedanta's maya-Brahman distinction.

What is genuinely new is the *combination*: accepting Bohr's complementarity while insisting on Einstein's realism, grounding Kantian limits in physical structure while claiming knowledge of noumenal character, embracing Cusanian learned ignorance while grounding it in a concrete physical mechanism — the Gödel $\to$ Wolpert $\to$ causal partition chain described above. The traditions reveal that this combination creates a productive philosophical tension — the paper claims to know the character of what it proves unknowable. Whether this tension is a contradiction (as Hegel would insist), a residual Platonism (as Nietzsche would charge), or a legitimate achievement of learned ignorance (as Cusa would celebrate) may be the framework's deepest philosophical question.

Frequently Asked Questions

"Isn't this just another interpretation of quantum mechanics?" No. Interpretations (Copenhagen, Many-Worlds, Bohmian mechanics) accept the quantum formalism and disagree about what it *means*. This framework *derives* the quantum formalism from non-quantum premises and proves the derivation is the only possible one. It makes quantitative predictions — the value of $\hbar$ and dark energy evolution — that interpretations do not, and it accounts for the dark-sector concordance as an automatic consequence.

"Doesn't Bell's theorem rule out hidden variable theories?" Bell's theorem rules out *local* hidden variable theories satisfying a specific factorizability condition. This framework violates that factorizability — not through faster-than-light signals, but because P-indivisible joint dynamics don't permit the decomposition Bell's theorem assumes. The framework reproduces exactly Tsirelson's bound (the maximum quantum violation), no more and no less.

"If everything is deterministic underneath, where does randomness come from?" From ignorance. The total system is deterministic, but the observer can't access the hidden sector. Different hidden states compatible with the same visible state lead to different outcomes. The observer must assign probabilities — not because the universe is random, but because their information is incomplete.

"What about dark matter?" Dark matter is not a substance — it is the local shape of the observer's blind spot, the same entropy that appears as dark energy when uniform. Baryonic matter displaces boundary entropy; the Jacobson mechanism converts the gradient into curvature; the crossover acceleration $a_0 = cH/6$ and baryonic Tully-Fisher relation $v^4 = GM_B \cdot cH/6$ follow with no fitted parameters, at the G1 register and conditional on the covariant rung G3. The mechanism works across all scales — from galaxies through clusters (Coma matched to $< 1\%$) to the Bullet Cluster and the CMB acoustic peaks — as detailed in the *Predictions* section above.

"What about the Bullet Cluster?" The horizon's response time ($\sim H^{-1} \approx$ 14 Gyr) vastly exceeds the collision crossing time (~ 0.15 Gyr). The boundary entropy is frozen at the pre-collision configuration — centered on the galaxies, not the recently displaced gas. The observed lensing morphology follows without collisionless particles (see *Cluster scales and the Bullet Cluster* above).

"How does this relate to recent mainstream work on observers in physics?" A substantial body of work since 2022 has independently arrived at the foundational claim that observation cannot be treated as external to physics — the same starting point as this framework, reached by different routes. Chandrasekaran, Longo, Penington, and Witten (CLPW, 2022) showed that adding an observer to quantum field theory in a gravitational subregion produces a finite well-defined entropy where previously there was none. Maldacena (2024) showed that an unphysical phase in the de Sitter partition function — long understood to obstruct a clean thermodynamic interpretation of the cosmological horizon — cancels exactly when an observer with a clock is properly incorporated. Harlow, Usatyuk, and Zhao (2025) and the AAIL construction (2025) showed that the dimension of the Hilbert space of quantum gravity in a closed universe is determined by the observer's degrees of freedom rather than being an independent parameter. Slagle and Preskill (2022) explicitly constructed a model in which quantum mechanics emerges at the boundary of a classical lattice with stochastic dynamics in an extra spatial dimension — a logical demonstration that QM-from-classical is achievable.

The framework here converges with these programs on the foundational substance — observation matters; the algebra of observables depends on the observer; the partition between observer and observed is physically meaningful — while reaching that conclusion through a different route. CLPW, Maldacena, and HUZ all add observer degrees of freedom *to* an existing quantum-gravitational formalism. The framework here goes the other direction: it starts with a deterministic substratum and a partition (the observer and the rest), and derives both quantum mechanics and the gravitational sector as consequences of that structural setup. There is no extra spatial dimension as in Slagle-Preskill, and no fundamental stochasticity. The substratum is finite and deterministic;

the observer's incomplete access to it is what produces the appearance of quantum behavior. Among emergent-QM programs the framework is distinguished by carrying twenty-two classified quantitative predictions to data (seven of them parameter-free structural) — a comparable empirical record is not currently produced by the alternatives. The convergence on observer-essentiality from many independent directions is encouraging context; the framework's empirical record is what makes the case specifically for this realization.

"How does this relate to string theory specifically?" String theory is the dominant unification program in fundamental physics, with several decades of work on whether the Standard Model can be reproduced from compactifications of higher-dimensional string vacua. The framework's relationship to string theory is examined directly in `Structure`. The findings: at the substratum level, the two frameworks are not equivalent. The framework's substratum is a finite deterministic bijection; string theory's matrix-model formulations are continuum stochastic path-integrals with non-linear Yang-Mills interactions. These structural commitments are different in ways that cannot be bridged by simple discretization or limiting procedures.

But the relationship at the observable level is more interesting. A four-feature audit of partial-trace observation — Born rule, channel-level unitarity, P-indivisibility (non-Markovian marginal), commutant gauge-invariance — finds that three of four transport directly between the frameworks. Holographic effective field theory in AdS/CFT explicitly classifies modes by Markovianity index, providing a direct analog of the framework's P-indivisibility. AdS/CFT bulk reconstruction provides a direct analog of the framework's Stinespring dilation, with bulk operators encoded in boundary subregions through quantum-error-correcting-code structure. The shared content is the structural features of partial-trace observation — what observation extracts from any quantum state under partial trace, independent of the substratum-level machinery producing the state. The fourth feature (commutant gauge-invariance) transports as a pattern but not as a specific output: both frameworks produce gauge invariance via commutant restriction, but the SM gauge group is forced uniquely from cubic-group representation theory in the framework's account, while it depends on compactification choice in string-theoretic constructions.

`Structure` takes this finding as the basis for a refined structural-realism claim. There exist universality classes of embedded observers, characterized by the structural features of observation rather than by substratum specifics. The framework's predictions live at multiple levels of universality: partial-trace features (most universal — shared across all embedded-observer systems); gauge-group structure (intermediate — shared across SM-reproducing classes); the framework's specific predictions like the Cabibbo angle $1/(\pi\sqrt{2})$ (most class-specific — features of the framework's specific representative within its universality class). The methodological reframing: fundamental physics as identifying the universality class of our universe through observational features, with substratum descriptions being framework-specific representations rather than competing claims about ultimate reality. In this picture, the framework and string theory are not in competition — they are different representations within (potentially) the same broader universality class, with the framework providing computationally tractable substratum-level structure that produces the framework's quantitative predictions, and string theory providing the rich machinery for quantum gravity and holography that the framework's GR cluster recovers thermodynamically through different routes.

"Why does gravity 'see' the classical vacuum energy and not the quantum one?" Because the spacetime metric exists at the classical level, *before* the quantum description emerges. The quantum zero-point energy is a feature of the observer's compressed description. It's real for quantum experiments (the Casimir effect, the Lamb shift) but doesn't appear in the stress-energy tensor that governs curvature. The 10^{122} discrepancy is the information compression ratio — the entropy of the observer's blind spot.

"How can the paper claim reality is 'definite' if it's permanently inaccessible?" This is the paper's most philosophically contested thesis (see *Philosophical Lineage* above). The paper's defense is that the claim follows from the derivation's own logic: the definition posits deterministic dynamics on a phase space, and the theorem shows that quantum indeterminacy arises from tracing out part of that phase space — not from any indeterminacy in the underlying evolution. The "definiteness" is a consequence of the starting premises, not a speculative addition. Whether those premises are the right ones to start from is, of course, an open question — but within the framework, Claim 4 is a theorem, not an assumption.

"Doesn't holographic physics show that spacetime comes from entanglement?" This is probably the strongest objection to the framework's ordering — classical spacetime first, quantum mechanics second. The Ryu-Takayanagi formula says entanglement entropy equals boundary area divided by $4G$. Van Raamsdonk argued that reducing entanglement disconnects spacetime. Programs like ER=EPR and "it from qubit" read these results as evidence that quantum entanglement is prior to geometry.

The framework offers an alternative reading. If the quantum description is produced by tracing out over a geometric boundary, then *of course* its entanglement entropy is proportional to the boundary's area — the information content of the trace-out is set by the number of modes crossing the boundary, which scales with area. The Ryu-Takayanagi formula, on this account, isn't a hint that entanglement builds geometry; it's a consequence of the fact that geometry built the entanglement. The Bekenstein-Hawking entropy $S = A/(4\,l_p^2)$, which the paper derives, is exactly this statement.

The correlation between entanglement and geometry is real either way. The question is which direction the arrow of explanation points. The two orderings make different empirical predictions: if geometry emerges from entanglement, spacetime structure should break down at high energy; if quantum mechanics emerges from geometry, the quantum description should break down near the discreteness scale while the geometric substratum persists. The cumulative case for the geometry-first ordering is already substantial: it produces the observed vacuum energy, the Bekenstein-Hawking 1/4 factor, the value of $\hbar$, the dark-sector concordance, the running-vacuum form of dark energy ($\nu \approx 0$, observationally $\approx \Lambda$CDM), the MOND acceleration scale $a_0 = cH/6$, the baryonic Tully-Fisher relation, the SM gauge group with its coupling strengths, the declining rotation curves at high redshift, $\bar{\theta} = 0$, the Coma cluster mass, and the Bullet Cluster lensing morphology — twelve independent consequences matching observation. The quantum-first ordering produces the 10^{122} discrepancy, treats $\hbar$ and the 1/4 factor as unexplained inputs, and has no natural account of the dark sector.

The Decompression Algorithm

There is a way to state what the framework means that is sharper than anything in the formal paper.

An observer inside the universe receives incomplete data — the visible sector only. The characterization theorem proves that the accessible statistics are necessarily memory-bearing, and that processes in the correspondence's class admits — under the translation hypothesis (T) — exactly the quantum algorithm for prediction.

Everything in QM is a feature of the algorithm, not a feature of the underlying reality. The wave function is the algorithm's internal state variable — the bookkeeping device it uses to track what it knows and what it doesn't. Complex amplitudes are the algorithm's arithmetic — the specific number system the reconstruction requires. Interference is what happens when the algorithm combines two incomplete pathways and their bookkeeping entries partially cancel. Entanglement is the algorithm's encoding of correlations that were written into the hidden sector during preparation and haven't been read back yet. The Born rule is the algorithm's output format — the way it converts its internal state into predictions the observer can check.

None of these are properties of the deterministic substratum. In the substratum, there are no wave functions, no complex numbers, no interference, no entanglement. There are configurations evolving under a Hamiltonian flow. That's all. The entire apparatus of quantum mechanics — every textbook, every equation, every experiment — is what the decompression algorithm looks like from the inside.

Antimatter as algorithmic artifact

This reframing changes the understanding of specific phenomena. Consider antimatter.

The standard account: the Dirac equation, which combines quantum mechanics with special relativity, has solutions with both positive and negative energy. The negative-energy solutions correspond to antiparticles — particles with opposite charge and quantum numbers. Every particle must have an antiparticle, because CPT invariance (a theorem of any local Lorentz-invariant quantum field theory) requires it.

The framework's account: the trace-out over the hidden sector forces the reconstruction algorithm to use two-signed amplitudes. When the algorithm operates in a relativistic context — which it must, because the substratum's causal structure is relativistic — the two-signed amplitude structure becomes the two-signed energy structure of the Dirac equation. Negative-energy solutions aren't additional features of reality. They're what the algorithm requires for self-consistency when the incomplete data comes from a relativistic system.

The parallel to the coin-and-die model is direct. The intermediate propagator:

$$\Lambda(2,1) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

has entries of -1 — "anti-probabilities" that don't exist in the substratum (where every transition probability is between 0 and 1). These negative entries are the minimal-model ancestors of antimatter. They arise for the same reason: the algorithm can't reconstruct the observed dynamics without them. In the toy model, the -1 entries encode the fact that the die remembers and reverses the coin flip — information backflow that no positive-entry matrix can describe. In the relativistic case, the negative-energy solutions encode the fact that the relativistic trace-out requires a doubled Hilbert space to remain self-consistent.

In both cases: the substratum has no doubling. The algorithm demands one.

Nothing in quantum mechanics explains a fundamental phenomenon

This is the deepest implication of the framework, and it is worth stating plainly.

The wave function does not explain what an electron is doing between measurements. It encodes what the algorithm computes given the observer's data. Interference does not explain why particles behave like waves. It reflects the algorithm's method of combining incomplete information. Entanglement does not explain a mysterious connection between distant particles. It reflects correlations stored in the hidden sector that the algorithm must track but cannot directly access. Antimatter does not explain a second kind of fundamental stuff. It reflects the algorithm's need for two-signed amplitudes in a relativistic context.

Every quantum phenomenon is an output of the decompression algorithm. The only fundamental phenomenon is the compression itself: the observer cannot see past the horizon, and the characterization theorem dictates the unique form of the resulting reconstruction.

This doesn't make quantum phenomena less real. Temperature is also an emergent phenomenon — a single molecule has no temperature. But temperature boils water, drives engines, and burns skin. Its emergence from statistical mechanics doesn't diminish its causal power within the emergent description. The same holds for every quantum phenomenon: they are causally potent, experimentally real, and the only physics accessible to an embedded observer. But they are features of the observer's necessary algorithm, not features of the universe's fundamental structure.

This resolves what is arguably the oldest open question in quantum foundations: *is the wave function real?* Since 1926, physics has been split between epistemic interpretations (the wave function is just a bookkeeping device tracking the observer's knowledge — it doesn't correspond to anything physical) and ontic interpretations (the wave function is a real physical entity — a field on configuration space, or a branching structure, or a guiding wave). The framework shows that both sides share a hidden assumption: that "real" means "fundamental." The epistemic camp says the wave function isn't fundamental, therefore it isn't real. The ontic camp says it's real, therefore it must be fundamental. Both are wrong. The wave function is real *and* not fundamental — in exactly the same way as antimatter, temperature, and every other emergent phenomenon. Within the bounded projection — which is all an embedded observer will ever have access to — the wave function, the positron, the interference pattern, and the electromagnetic field all have exactly the same ontological status. They are all outputs of the decompression algorithm. They are all experimentally real. And none of them exist in the substratum.

The analogy to data compression is precise. Lossy compression (like JPEG or MP3) discards information and reconstructs an approximation. The reconstruction has artifacts — ringing near edges in JPEG, pre-echo before transients in MP3. These artifacts are not in the original signal. They are features of the reconstruction algorithm applied to incomplete data.

Quantum mechanics is the reconstruction. The 10^{122} discarded bits (the hidden sector) are the lost information. Wave functions, interference, entanglement, and antimatter are the artifacts. They are real — as real as JPEG ringing is visible — but they are properties of the reconstruction, not of the original.

The difference from ordinary compression: with JPEG, you can access the original file. With the universe, you cannot. The reconstruction is all we will ever have. Its artifacts are our physics. And the characterization theorem proves that no other reconstruction is possible — this is the unique algorithm, and its artifacts are the unique artifacts. Quantum mechanics is not one possible decompression of incomplete cosmological data. It is the only one.

What This Means

The search for a "theory of everything" that unifies quantum mechanics and general relativity has assumed that the two theories describe the same level of reality and must be reconciled there. This paper argues that assumption is wrong. The two theories operate at different levels — one fundamental, one emergent — and their apparent contradiction is the information-theoretic cost of being an observer trapped inside the system you're trying to describe.

The apparent conflict between QM and GR, the 10^{122} vacuum energy discrepancy, and the dark sector are three faces of the same fact. The first is the existence of the trace-out. The second is its information compression ratio. The third is its gravitational occlusion fraction. All three are mandatory consequences of observational incompleteness — and all three match what we observe.

The universe is not broken. We are observing it from within.

Corroboration: The Rigidity Test

A natural objection to any framework this sweeping is: maybe it only works because it was built to work. Maybe the definition was chosen to produce QM, and the cosmological application was chosen because it fits. A flexible framework that can accommodate anything predicts nothing.

The companion papers `Substratum` and `SM` test this by asking a question the foundational paper doesn't address: if you build a concrete system satisfying the definition, does the framework constrain the dynamics? And if so, does the constrained dynamics produce anything beyond QM — something the framework wasn't designed to deliver?

It does, on both counts.

The dynamics is unique. Among all second-order reversible nearest-neighbor dynamics on a lattice, the requirements of center independence (necessary for emergent QM), spatial isotropy, and linearity select exactly one: the discrete wave equation. This holds for any alphabet size and any dimension. Center independence is necessary because center-dependent rules allow the visible sector to partially predict itself, suppressing the information backflow that produces the quantum phenomenology — an effect proven analytically on the full lattice via an information-screening mechanism. Linearity is selected by three independent criteria: it gives maximum propagation speed, it uniquely maximizes P-indivisibility among linear rules, and it is the only choice for which horizons are in thermal equilibrium (nonlinear dispersion breaks the Unruh effect).

The dynamics passes seven independent checks. The wave equation — selected solely by the QM requirement plus symmetry — turns out to produce: (1) non-Markovian reduced dynamics (emergent QM), (2) a causal structure with the correct spacetime dimension, (3) the gap equation for $\hbar$, (4) entanglement entropy proportional to boundary area, (5) a Lorentz-invariant dispersion relation at leading order, with linear Lorentz-invariance violation forbidden at the substrate level (the cubic-lattice dispersion admits no $\mathcal{O}(k^3)$ term) and the leading correction fixed at $\omega^2 = k^2[1-(4/45)(E/M_{\rm Pl})^2+...]$ — subluminal, with the 4/45 coefficient a specific prediction, not a free parameter, (6) the correct Unruh temperature at horizons, and (7) all inputs to Jacobson's thermodynamic derivation of Einstein's field equations. Each of these is an independent check that could have come out wrong — the entropy could have scaled with volume, the dispersion could have been non-relativistic or admitted linear LIV, the horizon state could have been non-thermal. None of these failures occurs.

No free parameters. The wave equation is selected, not chosen. The lattice spacing is fixed by the gap equation. The entropy coefficient is determined by thermal matching. The dispersion relation is an algebraic identity. There is nothing to tune.

This is what rigidity looks like. A framework that produces the quantum description from one set of arguments and then, without modification, produces the inputs for GR from a completely different set of arguments — lattice dynamics, dispersion relations, entanglement Hamiltonians — is making a structural claim that goes well beyond the original derivation. Each passed check is a point where the framework could have been falsified and wasn't. Seven passed checks with no free parameters is not proof, but it is the kind of evidence that distinguishes a framework describing something real from one that was engineered to fit.

What Is the Lattice?

The rigidity test proves that the wave equation on a lattice produces both QM and GR. But what *is* the lattice? Is the universe literally a grid of cells at the Planck scale? What would the grid be made of? What would it sit in?

The companion papers `Substratum` and `SM` address these questions head-on — and also derive the Standard Model's structure from the lattice dynamics. Their answer to the ontology question: the lattice is not a physical object. It is the *coupling structure* of the dynamics — the pattern of which degrees of freedom affect which others.

The minimal structure. The paper audits every assumption in the framework and identifies which are necessary for the theorems and which are artifacts of the particular construction. The result: six structural properties carry the emergent-QM, dimensional, and gravitational content — deterministic bijectivity, finite boundary entropy, bounded coupling degree, statistical isotropy, non-trivial partition coupling, and slow-bath capacity. For *that* content the specific alphabet size q and the wave equation are derived or irrelevant. The

Standard Model gauge group and the quantitative predictions are the exception: they additionally require the simple-cubic Bravais structure (the multiplicities (3,2,1) come from the octahedral point group), which is an empirically-selected input forced by the observed SU(3) factor, not derivable from statistical isotropy alone. For the emergent-QM core the minimal object is not a lattice; for the Standard Model it is the cubic lattice specifically. It is a triple: a finite set S, a bijection φ, and a partition V.

Space as coupling graph. Given any bijection on a finite set, you can ask: which components of the state affect which others? The answer defines a graph — sites are vertices, and two sites are connected if the dynamics couples them. This graph is not drawn on a pre-existing space. It IS space, at the fundamental level. Distance means "how many coupling steps apart." Area means "how many edges cross the boundary." The area law, the dispersion relation, the Myrheim-Meyer dimension — every spatial property used in the derivation chain is a property of this coupling graph.

This dissolves the container problem. The lattice doesn't sit in anything. There is no ambient manifold. The coupling graph is an abstract mathematical structure — like a number or a group — and any spatial embedding we draw is a representation for our convenience, not a physical fact. Asking "what does the lattice sit in?" is like asking what the number 7 sits in. It's a category error.

The alphabet is a gauge freedom. The companion papers prove every result for any alphabet size q $\geq$ 2. The gap equation contains no q. The Bekenstein-Hawking formula contains no q. The dispersion relation is valid for any q. No experiment could measure q, even in principle. This makes q a gauge freedom — a choice of mathematical description, like choosing a coordinate system or a gauge for the electromagnetic potential. The physical content is the coupling structure, not the microscopic state space.

Connection to causal set theory. The bijection's coupling graph, extended in time, generates a causal partial order: event A precedes event B if B is within A's future coupling light cone. This partial order is a causal set in the sense of Bombelli, Lee, Meyer, and Sorkin — the starting point of causal set theory, one of the established approaches to quantum gravity. Causal set theory has always had a specific gap: it postulates a causal order but lacks a deterministic dynamics yielding the quantum and gravitational descriptions. The OI framework provides exactly this. Sorkin's slogan is "Order + Number = Geometry." The OI version is "Bijection + Locality = QM + GR."

The hierarchy of physics. If the fundamental object is (S, φ) — a finite set and a bijection — then every concept in physics is a different aspect of this pair:

- φ is the dynamical law — the complete rule mapping states to states.
- **Space** is the coupling structure of φ — which degrees of freedom affect which others. It is determined by φ as the factorization minimizing coupling degree. Space is not a container; it is a relationship.

- **Matter** is the state — the values assigned to the degrees of freedom. A particle is a localized pattern that propagates through the coupling graph. Space and matter are the graph topology and the graph coloring; both come from (S, φ).
- **Energy** is the rate of change of the state under iteration of φ. A high-energy excitation changes rapidly from step to step; the vacuum changes least. Energy is not a substance — it is a measure of how fast a pattern evolves.
- **Time** is the iteration itself — the stepping from one state to the next. There is no continuous time at the fundamental level.
- **Quantum mechanics** is the observer's compressed view of V. It exists because the observer must marginalize over the hidden sector.
- **Gravity** is the thermodynamic limit of the coupling structure — the macroscopic behavior of bounded-degree graphs with area-law entropy.
- **Conservation laws** are emergent. Bijectivity of φ preserves state-space volume (information conservation). Energy conservation is what this looks like in the emergent quantum description (Noether's theorem). Momentum conservation reflects the symmetry of the coupling graph.

None of these are independent substances. They are the same structure viewed at different scales or from different angles. The framework does not unify them by reducing them to a common material. It unifies them by showing they were never separate.

What remains. The structural reading leaves little genuinely open. The spatial dimensionality $d = 3$ is derived by three independent self-consistency filters (propagating gravity, stable matter, and the boundary entropy concordance), with renormalizability of the emergent Yang-Mills coupling as a downstream consequence rather than an independent filter. Background independence is achieved through state-dependent bijections, with the discrete Einstein equation identified as the Ollivier-Ricci curvature condition. The observer is proved to be generic: any small subgraph of any large bounded-degree bijection satisfies the conditions for emergent QM. The fundamental object is (S, φ) — a finite set and a bijection. The partition V, the dimension, and the laws of physics are all derived. What remains undetermined are the 18+ parameter values of the Standard Model, which depend on the specific bijection φ — analogous to how Einstein's equations don't determine the mass of Jupiter.

But the central claim stands: the fundamental object is (S, φ) — a finite set and a bijection. The observer, the dimension, and the laws of physics are all emergent.

Why These Particles?

The first two companion papers establish the framework's identification of quantum mechanics with embedded observation and its derivation of general relativity. But quantum mechanics is a *framework*, not a *theory*. It tells the observer to use Hilbert spaces, unitary evolution, and the Born rule — but it's compatible with infinitely many different quantum field theories. You could have quantum mechanics with an SU(7) gauge group, or with 15 generations of fermions, or with no gauge fields at all. Identifying QM with embedded observation answers the question "what kind of probability does the observer use?" It doesn't answer "what particles exist?" or "what forces act between them?"

The Standard Model of particle physics — quarks, leptons, the strong and weak nuclear forces, electromagnetism, the Higgs boson — has been confirmed to extraordinary precision. But its *structure* has always been taken as empirical. We observe three generations, SU(3) × SU(2) × U(1), specific hypercharge assignments, and we accept them as given. Nobody has explained *why* these particles and not others.

`SM` asks whether the specific lattice dynamics selected by the QM and GR requirements — the wave equation on a d = 3 hypercubic lattice with checkerboard partition — determines which quantum field theory the observer sees. The answer is yes — almost completely.

Fermions from the wave equation. The wave equation selected by the QM requirement (the discrete lattice Klein-Gordon equation) has a well-known mathematical property: it factors into first-order operators called staggered Dirac operators. This factorization, discovered by Susskind in 1977, means the *same dynamics* that produces bosonic waves also describes fermionic matter. Fermions are not added to the framework. They *are* the framework, seen from a different mathematical angle.

The factorization produces a specific structure: on a three-dimensional lattice, the staggered construction yields exactly 4 "tastes" — independent species of fermions. These decompose as 1 + 3 under the cubic symmetry of the lattice: one singlet and one triplet. The triplet count equals the spatial dimension $d = 3$. Since $d = 3$ is derived in `Substratum`, the three-generation structure of the Standard Model — the fact that there are three copies of each type of quark and lepton — is traced back to the dimensionality of space.

Numerically, the singlet and triplet are not just distinguished by symmetry — they have quantitatively different coupling strengths. The singlet has coupling $|\mu| = 1$ (the maximum), while the three triplet members each have $|\mu| = 1/3$. The triplet members are exactly degenerate by cubic symmetry, providing a lattice-level origin for "generation symmetry."

The Higgs mechanism from one condition. The same center independence condition ($\alpha = 0$) that produces the quantum phenomenology's backflow structure also enforces exact chiral symmetry in the staggered fermions. This means fermion mass terms are forbidden in the fundamental Lagrangian. The *only* way to give fermions mass while preserving unitarity and renormalizability is the Higgs mechanism. One algebraic condition — center independence —

simultaneously produces the quantum phenomenology, chiral fermions, and the necessity of a Higgs boson.

Gauge structure from a matrix. The wave equation naturally generalizes to multiple components. Each lattice site carries a K-component vector instead of a single number. The selection criteria (center independence, isotropy, reversibility, linearity) uniquely produce the *matrix wave equation*, where a coupling matrix M is the sole new parameter.

This matrix M determines everything about the gauge structure. Its eigenvalue multiplicities give the gauge group — the group of transformations that leave M invariant (technically, its commutant in U(K)). Its eigenvalues give the mass spectrum. The gauge group and the mass spectrum are dual descriptions of the same matrix. This is a theorem, not an approximation.

Why K = 6. In three dimensions, each lattice site has 6 nearest neighbors (±x, ±y, ±z). If the internal components correspond to these link directions — the natural choice from the factorization principle, which says to decompose the per-site state space into its independent dynamical channels — then $K = 2d = 6$.

These 6 link directions form a representation of the cubic rotation group O. Standard character theory decomposes this representation into irreducibles:

$$6 = T_1(3) \oplus E(2) \oplus A_1(1)$$

The dimensions are 3, 2, and 1. By Schur's lemma, the coupling matrix M acts as an independent scalar on each irreducible subspace: $M = \text{diag}(\mu_c I_3, \mu_w I_2, \mu_y)$. The gauge group — the commutant of M — is therefore:

$$U(3) \times U(2) \times U(1) \supset SU(3) \times SU(2) \times U(1)$$

This is the Standard Model gauge group. It comes from the representation theory of the cubic lattice in three spatial dimensions. The three factors correspond to the three irreducible representations of O: the vector (T_1, dimension 3) gives color SU(3), the quadrupole (E, dimension 2) gives weak SU(2), and the scalar (A_1, dimension 1) gives hypercharge U(1).

Chiral coupling. The Standard Model has a peculiar feature: the weak force treats left-handed and right-handed particles differently. Left-handed particles form SU(2) doublets; right-handed particles are SU(2) singlets. No other known force has this asymmetry. Where does it come from?

In the OI framework, part of the answer is the partition itself — and part remains open, and is marked as such. The checkerboard partition that produces the memory-bearing dynamics quantum mechanics represents — even sites visible, odd sites hidden — coincides with the natural $\mathbb{Z}_2$ grading of staggered fermions. That grading, however, is the *product* of spin chirality and taste chirality ($\gamma_5 \otimes \xi_5$), not spin chirality alone: each sublattice carries both left- and right-handed components. What the partition does supply is a striking identity: on the visible sector, spin chirality and taste chirality become the same operator. The weak force's handedness then reduces to one sharp question — does the SU(2) embedding select a taste chirality? The minimal link-current embedding is now certified *not* to — it is taste-blind, exactly like color's (`hchi_selectivity_probes.py`, exact) — so the framework records the sharpened hypothesis H-χ': whether a non-minimal mechanism (a taste-asymmetric condensate structure entering the effective theory) restores selectivity. The honest summary: the substratum explains why a chirality-shaped grading exists, certifies that both the color and the minimal weak embeddings are blind to it, and leaves the weak force's specific handedness to the sharpened open mechanism.

Meanwhile, color SU(3) acts on the *internal* components (the K-component index at each site), which are preserved by the spatial trace-out. Both left and right fields carry color. SU(3) remains vector-like. The distinction: the grading lives in the spatial structure (touched by the trace-out); color lives in the internal structure (unaffected).

The strong CP problem, conditionally. QCD has a parameter $\bar{\theta}$ that violates CP symmetry. Experimentally, $|\bar{\theta}| < 10^{-10}$. Why is it so small? This is the strong CP problem. The standard solution invokes a new particle (the axion).

The OI framework addresses it without new particles, within the construction: T-invariance of the substratum wave equation gives $\bar{\theta} = 0$ at every scale of the emergent description — with four conditions standing between that result and the Standard Model's strong CP problem, none discharged ([SM §5.5]); no axion is invoked. The wave equation is exactly invariant under time reversal T. The trace-out is a spatial operation — it doesn't break T. Therefore the emergent QFT inherits T-invariance. The θ-term in QCD is T-odd, so T-invariance forces $\theta = 0$. The physical parameter $\bar{\theta} = \theta + \arg\det(Y_u Y_d)$ also vanishes because the visible-sector transition matrix satisfies detailed balance ($T_{ij} = T_{ji}$, proved and verified numerically), which constrains the Yukawa matrices to be real. The prediction, within the construction: $\bar{\theta} = 0$ with no axion. This is conditional rather than settled — the framework has still to exhibit a CP-violating weak sector inside the same construction, an emergent-level mechanism fixing $\bar{\theta}$, radiative stability against that sector, and a quantitative neutron-EDM number rather than an exact zero.

No empirical inputs. The hypercharge $U(1)_Y$ — traditionally an empirical input — is automatic: any $U(n)$ gauge group contains a U(1) factor, and anomaly cancellation uniquely determines the hypercharge assignments. The parity violation — also traditionally empirical — follows from the partition's chirality structure. The filter chain from (S, φ) to the Standard Model takes the spatial dimension d = 3 and the simple-cubic Bravais structure as empirically-selected inputs (the latter forced by the observed SU(3) factor); given those, the gauge group, the hypercharges, and the parity violation follow with no further empirical input. The hypercharge and parity results are genuine consequences; the cubic lattice on which the gauge derivation runs is an input, not a consequence of statistical isotropy alone.

Where the matter representations come from. A subtle point: the K = 6 internal space decomposes as a *direct sum* 3 + 2 + 1, but the SM quark doublet $\mathrm{Q}_L = (3, 2, +1/6)$ is a *tensor product* representation — it transforms simultaneously under both SU(3) and SU(2). This representation doesn't come from the lattice geometry. It comes from the *observer's existence*: the observer genericity theorem ([SM §8.2] Theorem 22) guarantees coupled, capacity-floored partitions generically, and the universal representation ([Main §3.4]) supplies every such observer's self-consistent quantum description, under (T). A self-consistent chiral QFT requires anomaly cancellation. Anomaly cancellation with SU(3) $\times$ SU(2) $\times$ U(1) uniquely determines the matter content per generation: Q_L, u_R, d_R, L_L, e_R with specific hypercharges. The lattice gives the gauge group and the generation count; the observer's self-consistency gives the representations. Three degenerate tastes then give three copies — three candidate matter sectors, which read as the three generations under H-spin' ([SM §4.7]).

What the framework does not determine. The coupling matrix M has three independent eigenvalues (μ_c, μ_w, μ_y), constrained by the max-speed condition to have $\mu_y = 1$. The remaining two eigenvalues set the strong and weak coupling strengths at the lattice scale. These, along with the Yukawa couplings (which determine fermion masses and mixing angles) and the Higgs potential parameters, are properties of the specific bijection φ — the specific "universe" the observer inhabits. The companion papers determine the *form* of the laws but not the *constants*. However, the universal induced coupling $1/\alpha_0 = 23.25$ IS structural — it depends on $N_f = 6$ and $T(R) = 1/2$, both determined by the lattice, not on the eigenvalues of M. Combined with non-perturbative gauge self-energy corrections and SM RG running, this reproduces all three observed gauge couplings at M_Z ([SM §6]). [SM §7] extends this further: the $d = 3$ lattice geometry organizes twenty-two SM observables — fermion mass ratios, CKM and PMNS mixing angles, the Higgs mass — of which seven are parameter-free structural retrodictions and the remainder are layered, chained, or fitted, each classified explicitly in [SM §7.6].

Predictions. The companion papers make seven structural predictions: (1) no additional gauge groups below the Planck scale; (2) no fundamental scalars beyond the Higgs; (3) no supersymmetric partners; (4) no fourth generation; (5) $\bar{\theta} = 0$ exactly, no axion; (6) flavor-sector CP phases in CKM and PMNS are solution-specific inputs (set by the basis-alignment of the specific φ) — compatible with but not derived by the framework, on the same footing as the fermion mass spectrum [SM §5.4]; (7) neutrinos are Majorana — the singlet taste is the Higgs sector (not a right-handed neutrino), so neutrinoless double beta decay should be observed. The taste-breaking mechanism ([SM §8.4]) further predicts: normal mass ordering ($m_3 > m_2 > m_1$), large PMNS mixing angles (from the structural mismatch between Dirac and Majorana mass matrices under the cubic group), and a hierarchical spectrum with Σm_ν near the oscillation minimum (≈ 0.059 eV). DESI DR2+CMB data ($\Sigma m_\nu < 0.064$ eV, normal ordering preferred) are consistent with all of these.

Combined with the earlier papers' predictions — dynamical dark energy in Type II RVM structural form (functional form derived from boundary-entropy architecture; theorem-level magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}| \sim 10^{-32}$, consistent with Bertini 2025's value-consistent-with-zero measurement at 2σ), dark matter from galaxies through clusters to the Bullet Cluster, gauge couplings at M_Z — and [SM §7]'s twenty-two zero-parameter quantitative predictions (including $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ matching the JUNO-era global fit at 0.07σ — a retrodiction), the framework makes a suite of falsifiable predictions that no competing theory makes jointly.

The Full Picture

The framework now covers the three pillars of modern physics and their consequences:

Pillar	What is derived	Source paper
Quantum mechanics	Emerges from embedded observation	Main
General relativity	Emerges from lattice dynamics passing 7 checks	GR + Substratum
Standard Model structure	SU(3) × SU(2) × U(1), 3 generations, Higgs, chiral coupling, $\bar{\theta} = 0$	SM
Structural foundations	$d = 3$, coupling graph ontology, q-gauge, background independence	Substratum
Chemistry → life	Orbitals, periodic table, carbon, water, chirality, autocatalysis, origin of life, evolution	Complexity

The starting point for all of it: a finite set, a bijection, and a partition.

From physics to life. The derived structure supplies the preconditions for organic chemistry with no fitted parameters: $d = 3$ determines orbital structure, the periodic table, carbon's tetrahedral bonding, water's solvent properties, and a thermal window for dynamic chemistry. The partition produces parity violation, which selects L-amino acids through autocatalytic amplification of the parity-violating energy difference. Combinatorial diversity exceeds Kauffman's autocatalytic threshold by 10^{56}. The origin of life is identified as the first molecular system satisfying C1–C4 — the same conditions that diagnose an accessible non-Markovian realization at the cosmological scale. RNA is the unique molecule that is simultaneously template (C2, C3) and catalyst (C1), making the RNA world a structural prediction rather than a historical hypothesis. C1–C4 systems grow exponentially while ordinary chemistry grows linearly, so the transition is inevitable and irreversible. Darwinian evolution follows from imperfect template replication. The viable parameter fraction is ~16%; fine-tuning and the anthropic principle dissolve.

The framework does not claim to be a complete theory of everything. It does not determine the 18+ parameters of the Standard Model — these depend on the specific bijection φ, just as the mass distribution of the universe depends on initial conditions, not on Einstein's equations. What it does claim — and what the derivation chain supports — is that the *structure* of the known laws of physics is not arbitrary. It is what mathematical consistency looks like when a finite deterministic system is observed from within. The fundamental object is (S, φ): a finite set and a bijection. Everything else — the observer, the dimension, quantum mechanics, general relativity, and the Standard Model — is emergent.

What Is (S, φ)?

The papers derive what (S, φ) *produces*. But what kind of thing *is* it?

Storage and memory

S is the set of all distinguishable states. Its physical meaning is *finite capacity*: there are |S| configurations that can be told apart, carrying $\log_2|S|$ bits of information. The finiteness is not a regularization — it's load-bearing. Finite sets recur (φ^N = id), recurrence produces P-indivisibility, and the imported correspondence supplies the quantum representation — for its whole class, with P-indivisibility fixing whether the represented dynamics is nontrivially indivisible. Strictly, only the visible sector and boundary layer need to be finite — the deep hidden sector may be infinite, because the accessible-timescale backflow lemma (Main, §2.3) establishes P-indivisibility without recurrence, and the boundary-only dependence lemma (GR, §3.2) ensures the deep sector's contribution is suppressed by $\tau_S/\tau_B \sim 10^{-32}$.

φ is a bijection: every state has exactly one predecessor and exactly one successor. Its physical meaning is *perfect memory*. Information is never created, never destroyed. The past is always recoverable from the present, because φ^{-1} exists. A non-bijective map would send two states to the same successor, erasing the distinction between them. φ preserves all distinctions, permanently.

Together, (S, φ) is a *finite lossless memory*: a register with finite capacity whose contents are updated without loss at each step. The partition V defines the observer's *access* — both read and write. The observer reads the visible sector (measuring x) and writes to the hidden sector (each visible-sector operation imprints correlations on H through coupling). The hidden sector stores those writes (C2: memory persistence — the record survives the window, whether by slow evolution or by conserved fast dynamics) and returns them on subsequent reads — producing the information backflow that P-indivisibility diagnoses — the memory-bearing sector where the distinctly quantum phenomenology lives; representability itself is universal.

The framework's results all have a clean interpretation in this language. The conditions C1–C4 are access conditions on the memory: C1 means the readable and unreadable parts are dynamically linked (read-write coupling is active), and C4 means the visible past is re-expressed through the hidden sector — in the physical realization, what is written is read back. C2 means the unreadable part retains its contents between reads. C3 means the unreadable part is large enough to store the full interaction history without overflow.

Quantum mechanics is the representation of this *read-write cycle* at the finite observable-law layer — not passive observation of a static memory, but the description of a finite subsystem that both reads from and writes to a register it cannot fully access. The fixed-basis Schrödinger/unitary interpolation and the quadratic Born readout are already part of the exact equivalent representation ($S \iff D \iff Q_{\mathrm{fb}}$, [Main §3.4]); they are not a separate exponent-selection problem. The unresolved question is whether the whole coherent intervention/composite family lifts into one standard local quantum operational theory. Interference in the mechanism picture is write-then-read: information deposited in hidden storage during one transition is retrieved on a later transition, partially cancelling or reinforcing the transition probabilities. The 10^{122} cosmological constant discrepancy is the compression ratio between total storage and readable storage. The dark sector is the gravitational weight of the unreadable storage. The Bekenstein-Hawking entropy is the storage capacity of the partition boundary. $\hbar$ is the conversion factor between storage geometry and read-write statistics.

An immediate objection: "Memory made of *what*?" In everyday life, memory requires a physical substrate — silicon, magnetic domains, neurons. The answer: (S, φ) is a *complete description* of reality — it determines every observable, with nothing left over that any measurement could detect. S is the totality of distinguishable configurations; φ is the structure of dynamical dependence. Space, time, and matter are all derived from (S, φ), so they can't appear in its definition without circularity. Whether (S, φ) *is* reality or *describes* reality is a question the framework proves to be gauge — undecidable by any experiment. The "memory" is a complete mathematical description from which everything observable is derived. Asking what it's made of presupposes a deeper level that no measurement can access — like asking what the number 7 is made of.

Computation from within

A Turing machine has a tape (storage), a head (read/write access), and a transition function (update rule). Under effective finiteness — where the visible sector and boundary layer are finite but the deep hidden sector may be infinite (Main, §4.3) — the correspondence between (S, φ, V) and a reversible Turing machine is not an analogy. It is structural.

The visible sector V is the head: a finite-state subsystem with bounded local coupling. The hidden sector H is the tape: a memory register that stores correlations written by past interactions. φ is a reversible transition function: no erasure, no halting, no information loss. The conditions C1–C4 characterize this architecture: C1 (coupling) is the read/write mechanism, and C4 (readback) closes the loop: history-sensitive hidden mediation, realized cosmologically as a read-write cycle. C2 (memory persistence) says the tape retains its contents between head operations. C3 (capacity) says the tape is large enough to store the full interaction history.

So what's different? Not the finiteness — the tape can be infinite. Not the reversibility — that's a specialization (reversible Turing machines are a well-studied subclass, going back to Bennett and Fredkin-Toffoli). The one genuine difference is the *question asked*.

Turing asked: what can a machine compute for an external observer who feeds in input and reads output?

The framework asks: what does computation look like to a component of the machine that cannot read the full tape?

At the finite observable-law layer, the answer is an exact quantum **representation**. The head cannot access the tape's global state. It sees only local statistics — transition probabilities obtained by marginalizing over unread tape cells. With readback those statistics are non-Markovian and, on a fixed finite recurrent representative, globally indivisible. The fixed-basis Hilbert/unitary/Born-form description is exact ($S \iff D \iff Q_{\mathrm{fb}}$). The stronger statement that superposition, entanglement, and arbitrary coherent instruments all belong to one standard local quantum operational theory is the separate coherent/composite lifting problem, not something the tape analogy proves by itself.

People ask: "Is the universe a computer?" The framework's answer: the universe is a reversible computation. And physics — quantum mechanics, general relativity, the Standard Model — is what that computation looks like to a subsystem embedded within it, observing through a finite window.

The arrow of time

If the universe is a finite permutation, it must eventually cycle back: φ^N = id for some N. So where does the arrow of time come from? Why does entropy increase?

The substratum itself has no arrow of time. φ and φ^{-1} are equally valid. The total fine-grained entropy is exactly conserved. So the arrow, if there is one, has to come from somewhere else — specifically, from the observer.

A structural theorem (Main §4.6) establishes exactly that. Observers satisfying C1–C4 — the conditions that make their description non-Markovian, quantum for the source's finite-configuration class — cannot exist in the equilibrium phase of (S, φ). The reason is direct. C2 requires the written record to persist to readback — and in a thermal bath the record lives in correlations, so persistence needs the hidden sector's correlation time τ_B to exceed the observer's measurement timescale τ_S (the conserve-and-route alternative being unavailable in an equilibrated bath). In equilibrium, local correlations decay on thermal timescales (a picosecond or so at ordinary temperatures), while any physical observer's internal processing time is many orders of magnitude longer. So in equilibrium, C2 fails — and with it C4: a bath that thermalizes between steps erases the stored record before it can be read (the erasure lemma, [Main §2.3]). Since C4 is what the characterization requires, no observer satisfying the framework's definition can exist there.

This has two striking consequences. First, the "Boltzmann brain" problem dissolves at the level of definitions. A Boltzmann brain is by construction a local fluctuation out of a thermally equilibrated environment — that's what "fluctuation from equilibrium" means. Its hidden sector is therefore the thermally equilibrated bath around it, with τ_B bounded by thermal decorrelation — too short for C2 to hold, by six or more orders of magnitude for any brain-like observer. Boltzmann brains carry no memory-bearing quantum phenomenology — no persistent record, no backflow — though their statistics, like any finite law's, admit the representation; they aren't observers in the framework's sense. The exclusion is structural, not anthropic — it doesn't depend on any argument about "typical observers" or the trustworthiness of memory.

Second, the "past hypothesis" of cosmology gets replaced by a theorem. Every standard cosmology has a mystery at t=0: why was the early universe so low-entropy? Boltzmann, Penrose, Albert, and Carroll all treat this as a postulate that has to be assumed separately. The framework replaces the postulate with a derivation. You don't need to posit low initial entropy to explain why we see it; you need only the definition of observation, which by itself selects for the non-equilibrium phase. Observers exist where C1–C4 hold. C1–C4 don't hold in equilibrium. Therefore observers exist in non-equilibrium. No additional axiom required.

In our universe, the non-equilibrium phase is provided by the expanding cosmological horizon. The horizon's expansion sets $\tau_B \sim H^{-1}$ — about 10^{17} seconds — which is structurally (not thermally) generated and is vastly longer than any laboratory timescale. This is why we see the memory-bearing quantum phenomenology at all: we're in the one phase of the universe where the framework's observer-conditions can be met — representability itself is universal; what the phase selects is the nontrivial sector. Entropy increase, the Second Law, and the felt direction of time emerge as coarse-grained descriptions of the dynamics accessible to observers in this non-equilibrium phase. Like quantum mechanics itself, the direction of time is emergent — a consequence of observing a lossless cycle from within, during the phase when such observation is structurally possible.

The incompleteness family

This places the framework in a family of results where self-reference under finite resources produces rigid structure.

Gödel (1931): A sufficiently powerful formal system cannot prove all truths about itself. The unprovable statements aren't random — they have a precise structure (the Gödel sentence, the consistency statement, the arithmetical hierarchy).

Turing (1936): A universal computer cannot decide all questions about its own behavior. The undecidable problems aren't random — they have a precise structure (the halting problem, Turing degrees, the oracle hierarchy).

OI: An observer embedded in (S, φ) cannot access the complete state. The emergent description isn't random — it has a precise structure (unitary quantum mechanics with the Schrödinger equation, Born rule, and Bell violations).

The precise OI analog of the halting problem is: *can the observer determine the hidden-sector state h?* The hidden state has a definite value at every moment — the total system is deterministic. Different hidden states send the same visible state to different futures (the particle goes left or right, the spin is measured up or down). But the observer provably cannot determine h: multiple hidden states are compatible with any visible-sector history, and transition probabilities are averages over h. The structural consequence is quantum mechanics — just as the structural consequence of the halting problem is the architecture of computability theory.

The framework also identifies a distinct class of inaccessible quantities: the alphabet size q and the deep-sector cardinality |C_D|. These are not undecidable — they are *gauge*. Different values produce identical observables, so there is no fact of the matter to be inaccessible. The hidden state h is undecidable: it has a real answer the observer cannot reach. The cardinality |C_D| is gauge: there is nothing there to reach.

The common thread: a system with finite resources tries to completely model something it's part of, and the structural impossibility of doing so determines the *form* of what it produces instead. The limitation and the law are the same object viewed from two sides. The Turing connection is direct: the observer is the head of a reversible computation; the halting problem (the head cannot determine the tape's global state) and the QM emergence theorem (the head's read-write statistics are necessarily non-Markovian, and quantum-mechanical for processes in the correspondence's class under (T)) are two consequences of the same structural constraint — finite self-referential access to a lossless computation.

Gödel and Turing are usually read as negative results — limits on knowledge. The framework recasts the physical instance as *generative*. What is mathematically rigid here is the finite stochastic/reversible/fixed-basis quantum representation equivalence, together with the memory-bearing indivisible sector selected by readback. Calling the resulting description **full operational quantum mechanics** requires the additional coherent/composite lifting and locality conditions stated above; the stronger uniqueness claim is not being smuggled into the analogy.

Why does mathematics describe physics?

This reframes one of the oldest questions in the philosophy of science.

The standard view: mathematics *describes* physics. The physical world exists independently, and we discover that mathematical equations capture its behavior. Wigner called this the "unreasonable effectiveness of mathematics" — why should abstract structures developed by pure mathematicians turn out to describe the physical world?

Tegmark's answer: the physical world *is* a mathematical structure. But this is a metaphysical claim that can't be tested, and it doesn't explain *which* mathematical structure or *why* this particular one.

The OI framework provides a different answer, with a mechanism. Mathematics doesn't describe physics. Physics is what mathematics *looks like from inside.*

The mathematical structure is (S, φ) — a finite set and a bijection. An observer embedded in that structure, seeing only part of it through the partition V, produces a stochastic description of what it can see, and that finite observable law has an exact fixed-basis quantum representation. Not because QM was designed to describe nature, and not because nature is mysteriously mathematical, but because the structural constraints of embedded observation determine this representation class. The step from that exact representation class to the unique full local operational theory is the explicitly separated lifting problem.

The trace-out is the mechanism. It takes the complete mathematical structure — the full evolution matrix with all its eigenvalues, Jordan blocks, and nilpotent monodromy — and projects it onto what the observer can see: the semisimple spectral data, organized by the representation theory of the partition. Physics is this projection.

What gets erased? The nilpotent part — the Jordan blocks that exist over finite fields but vanish over the reals, the monodromy operator that the observer can't detect, the arithmetic data about eigenvalue orders in finite field extensions. This is genuine mathematical structure. It's *there* in the dynamics.

But the observer, looking through the keyhole of the partition, sees only the diagonalizable shadow.

So Wigner's puzzle dissolves, for physics. Mathematics isn't "unreasonably effective" at describing physics. Mathematics — at least the part of it that physics tracks — is the most complete description an embedded observer would have access to, with the reconstruction theorem guaranteeing the structural content is determinate modulo gauge. Physics is the observable part of that description — drawn by someone who can't see the whole thing. The observable part is effective because it's a faithful projection — the semisimple part genuinely captures the eigenvalue structure. But it's not the whole description. The full mathematical description has structure (the nilpotent monodromy) that no physical measurement an embedded observer can perform can reveal. Whether mathematics extends beyond the structural content that physics tracks, into domains an embedded physical observer cannot reach (and that may have substantive content in mathematical foundations or meta-mathematics), is a question the framework does not address.

But there's a stronger result. The reconstruction theorem proves the converse: starting from the observed physics, mathematical consistency forces you uniquely back to the equivalence class [(S, φ)]. The two descriptions determine each other.

The forward direction: start from (S, φ) and derive physics. That's the program's main result. The converse: start from the observed physics and reconstruct (S, φ). And the pieces are already in hand.

If you observe quantum mechanics with Bell violations on a finite system, the characterization theorem forces you into some (S, φ, V) — there must be a deterministic system with a partition. If your physics includes Einstein gravity via the thermodynamic route, the dynamics selection theorem pins the wave equation — it's the unique dynamics consistent with QM emergence, isotropy, and linearity. If you see SU(3) $\times$ SU(2) $\times$ U(1) with three generations, the gauge structure pins K = 6 with multiplicities (3, 2, 1) — the cubic group decomposition is unique. If $\bar{\theta} = 0$, the dynamics is T-invariant. If $\hbar$ has the specific value $c^3\varepsilon^2/(4G)$, the gap equation pins ε = 2l_p. And d = 3 is the unique self-consistent dimension.

Chain all of these together and you get: the observed physics determines a unique equivalence class [(S, φ)] — a finite set with a bijection, pinned up to the substratum gauge group $\mathcal{G}_{\text{sub}}$ [Substratum §4]. This gauge group has four generators: state relabeling (permuting the elements of S while conjugating φ), alphabet change (replacing $\mathbb{Z}/q\mathbb{Z}$ with any $\mathbb{Z}/q'\mathbb{Z}$), deep-sector enlargement (the cardinality of the deep hidden sector beyond the boundary layer is unobservable — it may even be infinite), and graph isomorphism up to statistical isotropy (any bounded-degree graph with d = 3 polynomial growth and statistical isotropy is gauge-equivalent to $\mathbb{Z}^3$ *for the emergent-QM, dimensional, and gravitational content*; for the Standard Model gauge group this generator is restricted to isomorphisms preserving the cubic octahedral structure, since the (3,2,1) decomposition requires it — see [Substratum §4, Theorem 24(iv)]). The reconstruction theorem establishes this list for the state-channel content via Stinespring uniqueness; the graph generator carries the cubic-structure qualification just noted.

This is exactly how physics always works. In general relativity, measurements determine an equivalence class of spacetime metrics modulo diffeomorphisms — not a unique coordinate system. In gauge theory, experiments determine an equivalence class of connections modulo gauge transformations — not a unique field configuration. Here, the observed physics determines an equivalence class of finite bijections modulo the OI gauge freedoms — not a unique (S, φ).

So the question "is mathematics reality, or does it just describe reality?" is not unanswered — it's provably unanswerable by any measurement. The observed physics and the mathematical structure determine each other uniquely up to gauge, under the reconstruction theorem's stated conditions. Any experiment that could distinguish "the universe is (S, φ)" from "the universe is described by (S, φ)" would need to detect something outside the equivalence class. But the equivalence class already accounts for everything observable.

This is the same structure as gauge invariance in electromagnetism. The potential A is underdetermined — only the field strength F is measurable. We don't agonize over whether A is "real." We identify the exact boundary between the physical (F) and the gauge (A). The OI framework identifies the same kind of boundary for the math-physics relationship: the physics is the equivalence class [(S, φ)]; the ontological status of (S, φ) is gauge. The boundary is a theorem.

This suggests a precise way to understand what physics is, *from the embedded observer's perspective.* The mathematical structure (S, φ) has two parts: the part accessible to an embedded observer (the equivalence class, the gauge-invariant content, the semisimple spectral data), and the part that isn't (the specific representative, the nilpotent monodromy, the choice of q). The first part is what we call physics. A suggestive name for it might be *observable mathematics* — the portion of mathematical structure that an embedded subsystem with partial read-write access can detect.

Under this definition, Wigner's puzzle dissolves *for physics* to the point of tautology. "Why is mathematics effective at describing physics?" becomes "Why is the structural content tracked by physics effective at describing what an embedded observer detects?" — which isn't a mystery. It's a structural identification. Within physics, the unreasonable effectiveness was never unreasonable; it was a category error, treating the structural content of physics and the structural content of the mathematics that captures it as separate domains when, modulo gauge, they coincide. Whether mathematics in its full scope (including content beyond what any embedded physical observer can detect) is itself a coherent extension of this identification is a question for mathematical foundations, not for physics. The framework establishes the physics-side identification; whether the identification extends absolutely is a different question the framework does not claim to settle.

And the incompleteness family extends one step further: Gödel showed that proof can't capture all truth. Turing showed that computation can't predict all behavior. The OI framework shows that observation can't capture all dynamics. In each case, what *can't* be captured has rigid mathematical structure, and what *is* captured is determined by the incompleteness itself. Physics — all of it, from quantum mechanics to the Standard Model — is the rigid structure of what embedded observation captures. Mathematics is larger. Physics is the view from inside.

This is a companion overview to the framework's five core papers: "The Incompleteness of Observation" (Maybaum, [Main]), which presents the foundational embedded-observation theorems with detailed derivations; "The Standard Model from a Cubic Lattice" (Maybaum, [SM]), which derives the Standard Model's gauge group, matter content, and chiral structure from lattice dynamics; "The Gravitational Sector from the Cosmological Horizon" (Maybaum, [GR]), which derives ħ, the Bekenstein-Hawking area law, and dark-sector phenomenology; "The Substratum: Reconstruction Theorem and Gauge Group" (Maybaum, [Substratum]), which addresses the ontological status of the lattice and proves the framework's reconstruction theorem; and "Hierarchical Structural Realism and Universality Classes of Embedded Observers" (Maybaum, [Structure]), which articulates the framework's two-dimensional hierarchical structure and develops its relationship to other unification programs. The philosophical lineage section draws on a systematic analysis mapping the framework's claims against the major traditions in Western, Eastern, and contemporary philosophy of science.

Appendix: Mathematical Summary Tables

Sections 1–2: What Each Piece Does

Component	What it establishes	What it uses
Definition + Lemmas 1–3	The physical setup (no QM assumed)	Nothing — these are starting points
Partition-relativity (§1.4)	Emergent description depends only on partition	Definition, Lemmas 2, 3
Emergent stochasticity (§2.1)	Determinism looks random to embedded observer	Definition, Lemmas 2, 3
P-indivisibility proof (§2.3)	The stochastic process is non-Markovian	Definition, Lemma 1 + C1

Component	What it establishes	What it uses
Accessible-timescale lemma (§2.3)	Non-Markovianity is observable, not just formal	C1–C3, plus C4 holding quantitatively: two histories with the same present whose next-step laws differ by $\delta > 0$ (then I $\geq$ $p_0\delta^2/\ln 2$)
Coin-and-die model (§2.4)	Concrete demonstration of all mechanisms	Definition + all conditions
Stochastic-quantum correspondence (§3.1)	every process in the source's class embeds in a unistochastic one; P-indivisibility makes it nontrivial	Barandes [10,11] or Stinespring (Appendix A)
Necessity proof (§3.3)	Accessible non-Markovian memory requires C1/C3/C4 in every faithful deterministic realization; C2's physical-timescale necessity conditional on the mixing hypothesis	Per-condition necessity ([Main §3.4])

Sections 3–6: The Logical Flow

Step	What it establishes	Key equation
§3.1: Stochastic-quantum correspondence	compressed ($\leq$ n^3) representation for the correspondence's class; existence is internal and universal ($S \iff D \iff Q_{\mathrm{fb}}$, [Main §3.4])	T_ij = ancilla-marginal Born rule (T = $\|U\|^2$ in the trivial-ancilla case)
§3.1: Phase-locking	Hamiltonian fixed up to shift, rephasing, and the antiunitary conjugate	Fourier analysis of T_ij(t)
§3.2: Bell analysis	Spatial OI supplies causal cones; quantum CHSH requires setting screening + indivisibility	causal-local class $\Rightarrow S_{\mathrm{CHSH}} \leq 2\sqrt{2}$; bare OI has PR negative control

Step	What it establishes	Key equation
§3.3: Necessity of C1	Accessible memory requires coupling	No coupling → permutation → P-divisible
§3.3: Necessity of C2	Physical memory requires record persistence — slow bath one mechanism, conditional on mixing	Fast-mixing bath → Markov → P-divisible
§3.3: Necessity of C3	Accessible memory requires capacity	I ≤ $\log_2$ m bounds memory
§3.4: Characterization theorem	non-Markovianity $\Longleftrightarrow$ C1, C3, C4 per horizon (C2 separate, conditional); representation equivalence $S \Longleftrightarrow D \Longleftrightarrow Q_{\mathrm{fb}}$ — internal, universal; universal hidden-memory theorem	Biconditional + internal permutation unitary (compressed form imported under (T))
§4: Cosmological horizon	The universe satisfies the definition	C1 ✓, C2 ✓, C3 ✓, C4 ✓
§5.1: Classical temperature	T_cl = $c^2\varepsilon^2\kappa/(8\pi G k$_B)	No $\hbar$
§5.2 Step 1: Uniqueness	$\hbar$ is determined, not free	Partition-relativity
§5.2 Step 2: Boundary-only	$\hbar$ depends only on c, G, ε	Deep sector frozen
§5.2 Step 3: Dimensional analysis	$\hbar = \beta\ c^3\varepsilon^2/G$	Unique combination
§5.2 Step 4: Thermal matching	T_cl = T_Q fixes β = 1/4	$\hbar = c^3\varepsilon^2/(4G)$
§5.3: D-gauge theorem	No phase ambiguity	Double-difference condition
§6: Self-consistency	ε = 2 l_p, S_BH = A/(4l_p^2)	$\varepsilon^2 = 4\hbar G/c^3$

The chain from definition to $\hbar$:

Definition → universal Q_{fb} representation (P-indivisibility marking the memory sector) → Hamiltonian with unknown $\hbar$ → thermal matching at the horizon → $\hbar = c^3\varepsilon^2/(4G)$ → ε = 2 l_p → S_BH = A/(4 l_p^2)

Every link is either a proof or a calculation. No link requires quantum mechanics as input — QM appears as output at step 3 and its parameters are determined by steps 4–6.

Appendix: Key Mathematical Subtleties

1. The gap equation is a gap equation, not a derivation from nothing.

The framework has one free parameter (ε). The gap equation relates ε to $\hbar$. You need to know one to determine the other. The predictive content lies not in the gap equation alone but in its consequences: the specific relationship $\hbar = \mathrm{c}^3\varepsilon^2/(4\mathrm{G})$ — rather than any other function of c, G, ε — produces the Bekenstein-Hawking formula with the exact factor 1/4 (§6), the CC dissolution with S_dS as the compression ratio (§7.3), and the Type II running-vacuum form of the dark energy (§8.1). Any alternative $\hbar(\varepsilon)$ would fail at least one of these checks. The situation is analogous to deriving the Schwarzschild metric with M as a free parameter: the derivation has genuine content (the functional form) even though one input is not determined from within.

2. The KMS condition on the lattice.

The thermal matching in Step 4 uses the KMS condition, which is proved for continuum QFT on curved spacetime. The emergent QFT from Part I is lattice-regularized with cutoff ε. Thermal periodicity on a lattice has corrections of order $(\varepsilon\kappa/\mathrm{c})^2$. For the cosmological horizon: $\varepsilon\kappa/\mathrm{c} = \varepsilon\mathrm{H}/\mathrm{c} \sim 10^{-61}$, so corrections are $\sim 10^{-122}$. Negligible, but a lattice-native proof of the KMS condition would strengthen the argument.

3. The D-gauge theorem requires genericity.

The phase-locking and D-gauge results assume non-degenerate spectrum, non-degenerate energy gaps, and non-vanishing overlaps (conditions G1–G3). These fail on a measure-zero set of Hamiltonians. For the cosmological realization, a stronger argument applies to G1 specifically: the genericity conditions concern the *effective* visible-sector Hamiltonian $\hat{H}_{\text{eff}}$ (after the trace-out), not the total Hamiltonian H_{tot}. The relevant distinction is whether a symmetry of H_{tot} preserves the visible/hidden factorization. The spatial symmetries an observer-centered partition breaks — translation, rotation about distant points, boost — are *factorization-breaking* (they map boundary visible sites to hidden sites across the cosmological horizon), so they have no action on the visible factor alone and the degeneracies they protect are lifted in $\hat{H}_{\text{eff}}$. Factorization-preserving symmetries $S_V \otimes S_H$, by contrast, descend to a visible symmetry S_V of $\hat{H}_{\text{eff}}$ (the uniform-bath trace-out is invariant under any unitary on the hidden factor); the visible-internal ones among these are the gauge symmetries of the emergent QFT, which correspond to physically equivalent states and don't affect the phase-locking argument. G2 (non-degenerate gaps) is not symmetry-protected and rests on genericity directly; it holds for generic $\hat{H}_{\text{eff}}$ and can fail only on a measure-zero set, with no known structural feature of the substratum forcing systematic failure (single-cycle hidden dynamics in particular does not, since the Hermitian $\hat{H}_{\text{eff}}$ — unlike the unitary cycle operator's equally-spaced phases — has unequally-spaced real eigenvalues and retains non-degenerate gaps after the trace-out).

4. Why doesn't the deep sector matter?

The boundary-only dependence lemma is crucial because without it, $\hbar$ might depend on the vast interior of the hidden sector — which we can't observe or constrain. The lemma shows the deep sector is frozen on observable timescales, so its details are irrelevant. $\hbar$ is a property of the boundary geometry, not of the bulk.

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